



Chapter 2

Understanding Data

"Torture the data, and it will confess to anything."

— Ronald Coase

Machine learning algorithms involve large datasets. Hence, it is necessary to understand the data and datasets before applying machine learning algorithms. This chapter aims to introduce the concepts necessary to understand data better.

Learning Objectives

- Understand data and types of data
- Know about data management
- Learn the basics of dataset
- Identify the data types
- Introduce basics of descriptive analytics
- Explore data visualization techniques
- Introduce univariate data analysis
- Understand basics of bivariate and multivariate data analysis

2.1 WHAT IS DATA?

Scan for information on 'Machine Learning and Importance of Linear Algebra', 'Matrices and Tensors', 'Sampling Techniques', 'Information Theory', 'Evaluation of Classifier Model' and 'Additional Examples'



All facts are data. In computer systems, bits encode facts present in numbers, text, images, audio, and video. Data can be directly human interpretable (such as numbers or texts) or diffused data such as images or video that can be interpreted only by a computer. Today, business organizations are accumulating vast and growing amounts of data of the order of gigabytes, tera bytes, exabytes. A byte is 8 bits. A bit is either 0 or 1. A kilo byte (KB) is 1024 bytes, one mega byte (MB) is approximately 1000 KB, one giga byte is approximately 1,000,000 KB, 1000 giga bytes is one tera byte and 1000000 tera bytes is one Exa byte.

Data is available in different data sources like flat files, databases, or data warehouses. It can either be an operational data or a non-operational data. Operational data is the one that is encountered in normal business procedures and processes. For example, daily sales data is operational data, on the other hand, non-operational data is the kind of data that is used for decision making.

Data by itself is meaningless. It has to be processed to generate any information. A string of bytes is meaningless. Only when a label is attached like height of students of a class, the data becomes meaningful. Processed data is called information that includes patterns, associations, or relationships among data. For example, sales data can be analyzed to extract information like which product was sold larger in the last quarter of the year.

Elements of Big Data

Data whose volume is less and can be stored and processed by a small-scale computer is called 'small data'. These data are collected from several sources, and integrated and processed by a small-scale computer. Big data, on the other hand, is a larger data whose volume is much larger than 'small data' and is characterized as follows:

1. Volume – Since there is a reduction in the cost of storing devices, there has been a tremendous growth of data. Small traditional data is measured in terms of gigabytes (GB) and terabytes (TB), but Big Data is measured in terms of petabytes (PB) and exabytes (EB). One exabyte is 1 million terabytes.
2. Velocity – The fast arrival speed of data and its increase in data volume is noted as velocity. The availability of IoT devices and Internet power ensures that the data is arriving at a faster rate. Velocity helps to understand the relative growth of big data and its accessibility by users, systems and applications.
3. Variety – The variety of Big Data includes:
 - Form – There are many forms of data. Data types range from text, graph, audio, video, to maps. There can be composite data too, where one media can have many other sources of data, for example, a video can have an audio song.
 - Function – These are data from various sources like human conversations, transaction records, and old archive data.
 - Source of data – This is the third aspect of variety. There are many sources of data. Broadly, the data source can be classified as open/public data, social media data and multimodal data. These are discussed in Section 2.3.1 of this chapter.

Some of the other forms of Vs that are often quoted in the literature as characteristics of Big data are:

4. Veracity of data – Veracity of data deals with aspects like conformity to the facts, truthfulness, believability, and confidence in data. There may be many sources of error such as technical errors, typographical errors, and human errors. So, veracity is one of the most important aspects of data.
5. Validity – Validity is the accuracy of the data for taking decisions or for any other goals that are needed by the given problem.
6. Value – Value is the characteristic of big data that indicates the value of the information that is extracted from the data and its influence on the decisions that are taken based on it.

Thus, these 6 Vs are helpful to characterize the big data. The data quality of the numeric attributes is determined by factors like precision, bias, and accuracy. Precision is defined as the closeness of repeated measurements. Often, standard deviation is used to measure the precision. Bias is a systematic result due to erroneous assumptions of the algorithms or procedures. Accuracy is the degree of measurement of errors that refers to the closeness of measurements to the true value of the quantity. Normally, the significant digits used to store and manipulate indicate the accuracy of the measurement.

2.1.1 Types of Data

In Big Data, there are three kinds of data. They are structured data, unstructured data, and semi-structured data.

Structured Data

In structured data, data is stored in an organized manner such as a database where it is available in the form of a table. The data can also be retrieved in an organized manner using tools like SQL.

The structured data frequently encountered in machine learning are listed below:

Record Data A dataset is a collection of measurements taken from a process. We have a collection of objects in a dataset and each object has a set of measurements. The measurements can be arranged in the form of a matrix. Rows in the matrix represent an object and can be called as entities, cases, or records. The columns of the dataset are called attributes, features, or fields. The table is filled with observed data. Also, it is better to note the general jargons that are associated with the dataset. Label is the term that is used to describe the individual observations.

Data Matrix It is a variation of the record type because it consists of numeric attributes. The standard matrix operations can be applied on these data. The data is thought of as points or vectors in the multidimensional space where every attribute is a dimension describing the object.

Graph Data It involves the relationships among objects. For example, a web page can refer to another web page. This can be modeled as a graph. The nodes are web pages and the hyperlink is an edge that connects the nodes.

Ordered Data Ordered data objects involve attributes that have an implicit order among them.

The examples of ordered data are:

1. **Temporal data** – It is the data whose attributes are associated with time. For example, the customer purchasing patterns during festival time is sequential data. Time series data is a special type of sequence data where the data is a series of measurements over time.
2. **Sequence data** – It is like sequential data but does not have time stamps. This data involves the sequence of words or letters. For example, DNA data is a sequence of four characters – A T G C.
3. **Spatial data** – It has attributes such as positions or areas. For example, maps are spatial data where the points are related by location.

Unstructured Data

Unstructured data includes video, image, and audio. It also includes textual documents, programs, and blog data. It is estimated that 80% of the data are unstructured data.

Semi-Structured Data

Semi-structured data are partially structured and partially unstructured. These include data like XML/JSON data, RSS feeds, and hierarchical data.

2.1.2 Data Storage and Representation

Once the dataset is assembled, it must be stored in a structure that is suitable for data analysis. The goal of data storage management is to make data available for analysis. There are different approaches to organize and manage data in storage files and systems from flat file to data warehouses. Some of them are listed below:

Flat Files These are the simplest and most commonly available data source. It is also the cheapest way of organizing the data. These flat files are the files where data is stored in plain ASCII or EBCDIC format. Minor changes of data in flat files affect the results of the data mining algorithms. Hence, flat file is suitable only for storing small dataset and not desirable if the dataset becomes larger.

Some of the popular spreadsheet formats are listed below:

- CSV files – CSV stands for comma-separated value files where the values are separated by commas. These are used by spreadsheet and database applications. The first row may have attributes and the rest of the rows represent the data.
- TSV files – TSV stands for Tab separated values files where values are separated by Tab.

Both CSV and TSV files are generic in nature and can be shared. There are many tools like Google Sheets and Microsoft Excel to process these files.

Database System It normally consists of database files and a database management system (DBMS). Database files contain original data and metadata. DBMS aims to manage data and improve operator performance by including various tools like database administrator, query processing, and transaction manager. A relational database consists of sets of tables. The tables have rows and columns. The columns represent the attributes and rows represent tuples. A tuple corresponds to either an object or a relationship between objects. A user can access and manipulate the data in the database using SQL.

Different types of databases are listed below:

1. A transactional database is a collection of transactional records. Each record is a transaction. A transaction may have a time stamp, identifier and a set of items, which may have links to other tables. Normally, transactional databases are created for performing associational analysis that indicates the correlation among the items.
2. Time-series database stores time related information like log files where data is associated with a time stamp. This data represents the sequences of data, which represent values or events obtained over a period (for example, hourly, weekly or yearly) or repeated time span. Observing sales of product continuously may yield a time-series data.

3. Spatial databases contain spatial information in a raster or vector format. Raster formats are either bitmaps or pixel maps. For example, images can be stored as a raster data. On the other hand, the vector format can be used to store maps as maps use basic geometric primitives like points, lines, polygons and so forth.

World Wide Web (WWW) It provides a diverse, worldwide online information source. The objective of data mining algorithms is to mine interesting patterns of information present in WWW.

XML (eXtensible Markup Language) It is both human and machine interpretable data format that can be used to represent data that needs to be shared across the platforms.

Data Stream It is dynamic data, which flows in and out of the observing environment. Typical characteristics of data stream are huge volume of data, dynamic, fixed order movement, and real-time constraints.

RSS (Really Simple Syndication) It is a format for sharing instant feeds across services.

JSON (JavaScript Object Notation) It is another useful data interchange format that is often used for many machine learning algorithms.

2.2 BIG DATA ANALYTICS AND TYPES OF ANALYTICS

The primary aim of data analysis is to assist business organizations to take decisions. For example, a business organization may want to know which is the fastest selling product, in order for them to market activities. Data analysis is an activity that takes the data and generates useful information and insights for assisting the organizations.

Data analysis and data analytics are terms that are used interchangeably to refer to the same concept. However, there is a subtle difference. Data analytics is a general term and data analysis is a part of it. Data analytics refers to the process of data collection, preprocessing and analysis. It deals with the complete cycle of data management. Data analysis is just analysis and is a part of data analytics. It takes historical data and does the analysis. Data analytics, instead, concentrates more on future and helps in prediction.

There are four types of data analytics:

1. Descriptive analytics
2. Diagnostic analytics
3. Predictive analytics
4. Prescriptive analytics

Descriptive Analytics It is about describing the main features of the data. After data collection is done, descriptive analytics deals with the collected data and quantifies it. It is often stated that analytics is essentially statistics. There are two aspects of statistics – Descriptive and Inference. Descriptive analytics only focuses on the description part of the data and not the inference part.

Diagnostic Analytics It deals with the question – ‘Why?’. This is also known as causal analysis, as it aims to find out the cause and effect of the events. For example, if a product is not selling, diagnostic analytics aims to find out the reason. There may be multiple reasons and associated effects are analyzed as part of it.

Predictive Analytics It deals with the future. It deals with the question – ‘What will happen in future given this data?’. This involves the application of algorithms to identify the patterns to predict the future. The entire course of machine learning is mostly about predictive analytics and forms the core of this book.

Prescriptive Analytics It is about the finding the best course of action for the business organizations. Prescriptive analytics goes beyond prediction and helps in decision making by giving a set of actions. It helps the organizations to plan better for the future and to mitigate the risks that are involved.

2.3 BIG DATA ANALYSIS FRAMEWORK

For performing data analytics, many frameworks are proposed. All proposed analytics frameworks have some common factors. Big data framework is a layered architecture. Such an architecture has many advantages such as genericness. A 4-layer architecture has the following layers:

1. Data connection layer
2. Data management layer
3. Data analytics later
4. Presentation layer

Data Connection Layer It has data ingestion mechanisms and data connectors. Data ingestion means taking raw data and importing it into appropriate data structures. It performs the tasks of ETL process. By ETL, it means extract, transform and load operations.

Data Management Layer It performs preprocessing of data. The purpose of this layer is to allow parallel execution of queries, and read, write and data management tasks. There may be many schemes that can be implemented by this layer such as data-in-place, where the data is not moved at all, or constructing data repositories such as data warehouses and pull data on-demand mechanisms.

Data Analytic Layer It has many functionalities such as statistical tests, machine learning algorithms to understand, and construction of machine learning models. This layer implements many model validation mechanisms too. The processing is done as shown in Box 2.1.

Box 2.1: Types of Processing

Cloud Computing

Cloud computing is an emerging technology which is basically a business service model or simply called as pay-per-usage model. The term ‘Cloud’ refers to the Internet that provides sharing of processing power, applications, storage and services. It offers different kinds of services such as Iaas, Paas, and SaaS.

SaaS (Software as a Service) enables users to access software applications from the cloud. Paas (Platform as a Service) provides users the platform to develop and run their applications. Iaas (Infrastructure as a Service) enables users to access the infrastructure required to run their applications, storage, operating systems, etc.

(Continued)

The cloud services can be deployed in four most commonly used deployment models such as Public Cloud, Private Cloud, Community Cloud, and Hybrid Cloud based on the service model, organization, geographic location, etc. The **Public Cloud** is accessible to the public and is owned by a vendor, who offers the services of the cloud to the users. **Private Cloud** is a privately-owned cloud where the user or an organization owns the cloud and only the user or employees of that organization have access to the cloud, thereby making data and transactions secure. In **Community Cloud**, the infrastructure is owned jointly by different organizations. The **Hybrid Cloud** is the combination of two or more cloud types. The characteristics of cloud computing are:

1. Shared Infrastructure – Sharing of physical services, storage, and networking capabilities
2. Dynamic Provisioning – Resources assigned dynamically, based on demands
3. Dynamic Scaling – Expansion and contraction of service capability
4. Network Access – Needs to be accessed across the internet
5. Utility-based Metering – Uses metering to provide reporting and billing information
6. Multitenancy – Serves multiple customers
7. Reliability – Customer reliable service

Grid Computing

Grid Computing is a parallel and distributed computing framework consisting of a network of computers offering a super computing service as a single virtual supercomputer. This high-performance computing is required to perform specialized tasks that require a high computing power and a single computer cannot provide enough computing resources. The grid computing model forms a grid by connecting tens of thousands of nodes as a cluster that runs on an operating system. In this model, the resources are pooled together and the load is shared across multiple nodes to accomplish a task more quickly. This grid is constructed by middleware software that evenly distributes the task to several nodes connected in the grid. The individual nodes perform the task independently and in parallel which are then integrated to complete the large-scale task. This model of computing is best suited for applications that are complex and can be computed in parallel.

H-Computing (High Performance Computing or HPC)

It enables to perform complex tasks at high speed. It aggregates computing power in such a way that provides much higher performance to solve complex problems in science, engineering, research or business. It leverages parallel processing techniques for solving complex computational problems. HPC system achieves this sustained performance through concurrent use of computing resources. An HPC system combines the computing power of thousands of compute nodes that work in parallel to complete tasks faster. The system comprises three key components called compute, network and storage. The architecture of HPC consists of compute servers that are networked together to form a cluster. Software programs are run in parallel on the servers in the cluster and are networked to the data storage to capture the output. These components work together to complete a task.

Presentation Layer It has mechanisms such as dashboards, and applications that display the results of analytical engines and machine learning algorithms.

Thus, the Big Data processing cycle involves data management that consists of the following steps.

1. Data collection
2. Data preprocessing
3. Applications of machine learning algorithm
4. Interpretation of results and visualization of machine learning algorithm

This is an iterative process and is carried out on a permanent basis to ensure that data is suitable for data mining.

Application and interpretation of machine learning algorithms constitute the basis for the rest of the book. So, primarily, data collection and data preprocessing are covered as part of this chapter.

The following section covers data collection in detail.

2.3.1 Data Collection

The first task of gathering datasets are the collection of data. It is often estimated that most of the time is spent for collection of good quality data. A good quality data yields a better result. It is often difficult to characterize a 'Good data'. 'Good data' is one that has the following properties:

1. Timeliness – The data should be relevant and not stale or obsolete data.
2. Relevancy – The data should be relevant and ready for the machine learning or data mining algorithms. All the necessary information should be available and there should be no bias in the data.
3. Knowledge about the data – The data should be understandable and interpretable, and should be self-sufficient for the required application as desired by the domain knowledge engineer.

Broadly, the data source can be classified as open/public data, social media data and multimodal data.

1. Open or public data source – It is a data source that does not have any stringent copyright rules or restrictions. Its data can be primarily used for many purposes. Government census data are good examples of open data:
 - Digital libraries that have huge amount of text data as well as document images
 - Scientific domains with a huge collection of experimental data like genomic data and biological data
 - Healthcare systems that use extensive databases like patient databases, health insurance data, doctors' information, and bioinformatics information
2. Social media – It is the data that is generated by various social media platforms like Twitter, Facebook, YouTube, and Instagram. An enormous amount of data is generated by these platforms.
3. Multimodal data – It includes data that involves many modes such as text, video, audio and mixed types. Some of them are listed below:

- Image archives contain larger image databases along with numeric and text data
 - The World Wide Web (WWW) has huge amount of data that is distributed on the Internet.
- These data are heterogeneous in nature.

2.3.2 Data Preprocessing

In real world, the available data is 'dirty'. By this word 'dirty', it means:

- Incomplete data
- Inaccurate data
- Outlier data
- Data with missing values
- Data with inconsistent values
- Duplicate data

Data preprocessing improves the quality of the data mining techniques. The raw data must be preprocessed to give accurate results. The process of detection and removal of errors in data is called data cleaning. Data wrangling means making the data processable for machine learning algorithms. Some of the data errors include human errors such as typographical errors or incorrect measurement and structural errors like improper data formats. Data errors can also arise from omission and duplication of attributes. Noise is a random component and involves distortion of a value or introduction of spurious objects. Often, the noise is used if the data is a spatial or temporal component. Certain deterministic distortions in the form of a streak are known as artifacts.

Consider, for example, the following patient Table 2.1. The 'bad' or 'dirty' data can be observed in this table.

Table 2.1: Illustration of 'Bad' Data

Patient ID	Name	Age	Date of Birth (DoB)	Fever	Salary
1.	John	21		Low	-1500
2.	Andre	36		High	Yes
3.	David	5	10/10/1980	Low	" "
4.	Raju	136		High	Yes

It can be observed that data like Salary = ' ' is incomplete data. The DoB of patients, John, Andre, and Raju, is the missing data. The age of David is recorded as '5' but his DoB indicates it is 10/10/1980. This is called inconsistent data.

Inconsistent data occurs due to problems in conversions, inconsistent formats, and difference in units. Salary for John is -1500. It cannot be less than '0'. It is an instance of noisy data. Outliers are data that exhibit the characteristics that are different from other data and have very unusual values. The age of Raju cannot be 136. It might be a typographical error. It is often required to distinguish between noise and outlier data.

Outliers may be legitimate data and sometimes are of interest to the data mining algorithms. These errors often come during data collection stage. These must be removed so that machine learning algorithms yield better results as the quality of results is determined by the quality of input data. This removal process is called data cleaning.

Missing Data Analysis

The primary data cleaning process is missing data analysis. Data cleaning routines attempt to fill up the missing values, smoothen the noise while identifying the outliers and correct the inconsistencies of the data. This enables data mining to avoid overfitting of the models.

The procedures that are given below can solve the problem of missing data:

1. Ignore the tuple – A tuple with missing data, especially the class label, is ignored. This method is not effective when the percentage of the missing values increases.
2. Fill in the values manually – Here, the domain expert can analyse the data tables and carry out the analysis and fill in the values manually. But, this is time consuming and may not be feasible for larger sets.
3. A global constant can be used to fill in the missing attributes. The missing values may be 'Unknown' or be 'Infinity'. But, some data mining results may give spurious results by analysing these labels.
4. The attribute value may be filled by the attribute value. Say, the average income can replace a missing value.
5. Use the attribute mean for all samples belonging to the same class. Here, the average value replaces the missing values of all tuples that fall in this group.
6. Use the most possible value to fill in the missing value. The most probable value can be obtained from other methods like classification and decision tree prediction.

Some of these methods introduce bias in the data. The filled value may not be correct and could be just an estimated value. Hence, the difference between the estimated and the original value is called an error or bias.

Removal of Noisy or Outlier Data

Noise is a random error or variance in a measured value. It can be removed by using binning, which is a method where the given data values are sorted and distributed into equal frequency bins. The bins are also called as buckets. The binning method then uses the neighbor values to smooth the noisy data.

Some of the techniques commonly used are 'smoothing by means' where the mean of the bin removes the values of the bins, 'smoothing by bin medians' where the bin median replaces the bin values, and 'smoothing by bin boundaries' where the bin value is replaced by the closest bin boundary. The maximum and minimum values are called bin boundaries. Binning methods may be used as a discretization technique. Example 2.1 illustrates this principle.

Example 2.1: Consider the following set: $S = \{12, 14, 19, 22, 24, 26, 28, 31, 34\}$. Apply various binning techniques and show the result.

Solution: By equal-frequency bin method, the data should be distributed across bins. Let us assume the bins of size 3, then the above data is distributed across the bins as shown below:

Bin 1 : 12, 14, 19

Bin 2 : 22, 24, 26

Bin 3 : 28, 31, 32

By smoothing bins method, the bins are replaced by the bin means. This method results in:

Bin 1 : 15, 15, 15

Bin 2 : 24, 24, 24

Bin 3 : 30.3, 30.3, 30.3

Using smoothing by bin boundaries method, the bins' values would be like:

Bin 1 : 12, 12, 19

Bin 2 : 22, 22, 26

Bin 3 : 28, 32, 32

As per the method, the minimum and maximum values of the bin are determined, and it serves as bin boundary and does not change. Rest of the values are transformed to the nearest value. It can be observed in Bin 1, the middle value 14 is compared with the boundary values 12 and 19 and changed to the closest value, that is 12. This process is repeated for all bins.

Data Integration and Data Transformations

Data integration involves routines that merge data from multiple sources into a single data source. So, this may lead to redundant data. The main goal of data integration is to detect and remove redundancies that arise from integration. Data transformation routines perform operations like normalization to improve the performance of the data mining algorithms. It is necessary to transform data so that it can be processed. This can be considered as a preliminary stage of data conditioning. Normalization is one such technique. In normalization, the attribute values are scaled to fit in a range (say 0–1) to improve the performance of the data mining algorithm. Often, in neural networks, these techniques are used. Some of the normalization procedures used are:

1. *Min-Max*
2. *z-Score*

Min-Max Procedure It is a normalization technique where each variable V is normalized by its difference with the minimum value divided by the range to a new range, say 0–1. Often, neural networks require this kind of normalization. The formula to implement this normalization is given as:

$$\text{min-max} = \frac{V - \text{min}}{\text{max} - \text{min}} \times (\text{new max} - \text{new min}) + \text{new min} \quad (2.1)$$

Here max-min is the range. Min and max are the minimum and maximum of the given data, new max and new min are the minimum and maximum of the target range, say 0 and 1.

Example 2.2: Consider the set: $V = \{88, 90, 92, 94\}$. Apply *Min-Max* procedure and map the marks to a new range 0–1.

Solution: The minimum of the list V is 88 and maximum is 94. The *new min* and *new max* are 0 and 1, respectively. The mapping can be done using Eq. (2.1) as:

For marks 88,

$$\text{min-max} = \frac{88 - 88}{94 - 88} \times (1 - 0) + 0 = 0$$

Similarly, other marks can be computed as follows:

For marks 90,

$$\text{min-max} = \frac{90 - 88}{94 - 88} \times (1 - 0) + 0 = 0.33$$

For marks 92,

$$\text{min-max} = \frac{92 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{4}{6} = 0.66$$

For marks 94,

$$\text{min-max} = \frac{94 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{6}{6} = 1$$

So, it can be observed that the marks {88, 90, 92, 94} are mapped to the new range {0, 0.33, 0.66, 1}.

Thus, the *Min-Max* normalization range is between 0 and 1.

z-Score Normalization This procedure works by taking the difference between the field value and mean value, and by scaling this difference by standard deviation of the attribute.

$$V^* = V - \mu/\sigma \quad (2.2)$$

Here, σ is the standard deviation of the list V and μ is the mean of the list V .

Example 2.3: Consider the mark list $V = \{10, 20, 30\}$, convert the marks to z-score.

Solution: The mean and Sample Standard deviation (σ) values of the list V are 20 and 10, respectively. So the z-scores of these marks are calculated using Eq. (2.2) as:

$$\text{z-score of 10} = \frac{10 - 20}{10} = -\frac{10}{10} = -1$$

$$\text{z-score of 20} = \frac{20 - 20}{10} = \frac{0}{10} = 0$$

$$\text{z-score of 30} = \frac{30 - 20}{10} = \frac{10}{10} = 1$$

Hence, the z-score of the marks 10, 20, 30 are -1, 0 and 1, respectively.

What is the use of z-scores? z-scores are used to detect outlier detection. If the data value z-score function is either less than -3 or greater than +3, then it is possibly an outlier. The major disadvantage of z-score function is that it is extremely sensitive to outliers as it is dependent on mean.

Data Reduction

Data reduction reduces data size but produces the same results. There are different ways in which data reduction can be carried out such as data aggregation, feature selection, and dimensionality reduction.

2.4 DESCRIPTIVE STATISTICS

Descriptive statistics is a branch of statistics that does dataset summarization. It is used to summarize and describe data. Descriptive statistics are just descriptive and do not go beyond that. In other words, descriptive statistics do not bother too much about machine learning algorithms and its functioning.

Data visualization is a branch of study that is useful for investigating the given data. Mainly, the plots are useful to explain and present data to customers.

Descriptive analytics and data visualization techniques help to understand the nature of the data, which further helps to determine the kinds of machine learning or data mining tasks that can be applied to the data. This step is often known as Exploratory Data Analysis (EDA). The focus of EDA is to understand the given data and to prepare it for machine learning algorithms. EDA includes descriptive statistics and data visualization.

Let us discuss descriptive statistics with the fundamental concepts of datatypes.

Dataset and Data Types

A dataset can be assumed to be a collection of data objects. The data objects may be records, points, vectors, patterns, events, cases, samples or observations. These records contain many attributes. An attribute can be defined as the property or characteristics of an object.

For example, consider the following database shown in sample Table 2.2.

Table 2.2: Sample Patient Table

Patient ID	Name	Age	Blood Test	Fever	Disease
1.	John	21	Negative	Low	No
2.	Andre	36	Positive	High	Yes

Every attribute should be associated with a value. This process is called measurement. The type of attribute determines the data types, often referred to as measurement scale types. The data types are shown in Figure 2.1.

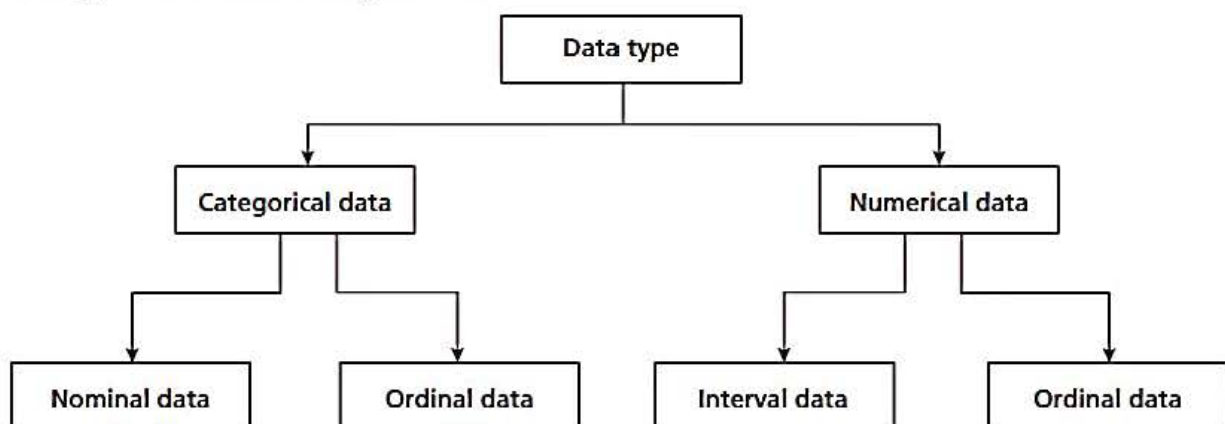


Figure 2.1: Types of Data

Broadly, data can be classified into two types:

1. Categorical or qualitative data
2. Numerical or quantitative data

Categorical or Qualitative Data The categorical data can be divided into two types. They are nominal type and ordinal type.

- **Nominal Data** – In Table 2.2, patient ID is nominal data. Nominal data are symbols and cannot be processed like a number. For example, the average of a patient ID does not make any statistical sense. Nominal data type provides only information but has no ordering among data. Only operations like ($=$, $\neq$) are meaningful for these data. For example, the patient ID can be checked for equality and nothing else.
- **Ordinal Data** – It provides enough information and has natural order. For example, Fever = {Low, Medium, High} is an ordinal data. Certainly, low is less than medium and medium is less than high, irrespective of the value. Any transformation can be applied to these data to get a new value.

Numeric or Quantitative Data It can be divided into two categories. They are interval type and ratio type.

- **Interval Data** – Interval data is a numeric data for which the differences between values are meaningful. For example, there is a difference between 30 degree and 40 degree. Only the permissible operations are $+$ and $-$.
- **Ratio Data** – For ratio data, both differences and ratio are meaningful. The difference between the ratio and interval data is the position of zero in the scale. For example, take the Centigrade-Fahrenheit conversion. The zeroes of both scales do not match. Hence, these are interval data.

Another way of classifying the data is to classify it as:

1. Discrete value data
2. Continuous data

Discrete Data This kind of data is recorded as integers. For example, the responses of the survey can be discrete data. Employee identification number such as 10001 is discrete data.

Continuous Data It can be fitted into a range and includes decimal point. For example, age is a continuous data. Though age appears to be discrete data, one may be 12.5 years old and it makes sense. Patient height and weight are all continuous data.

Third way of classifying the data is based on the number of variables used in the dataset. Based on that, the data can be classified as univariate data, bivariate data, and multivariate data. This is shown in Figure 2.2.

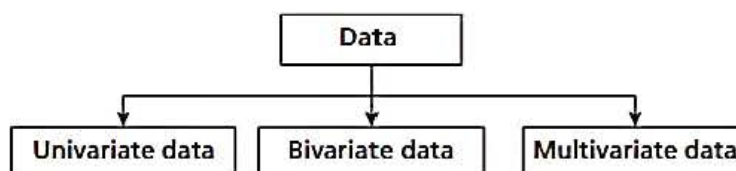


Figure 2.2: Types of Data Based on Variables

In case of univariate data, the dataset has only one variable. A variable is also called as category. Bivariate data indicates that the number of variables used are two and multivariate data uses three or more variables.

This chapter primarily deals with univariate data in detail with just an overview of bivariate and multivariate data.

2.5 UNIVARIATE DATA ANALYSIS AND VISUALIZATION

Scan for information on 'Measures of Frequency' and also for 'Additional Examples'



Univariate analysis is the simplest form of statistical analysis. As the name indicates, the dataset has only one variable. A variable can be called as a category. Univariate does not deal with cause or relationships. The aim of univariate analysis is to describe data and find patterns.

Univariate data description involves finding the frequency distributions, central tendency measures, dispersion or variation, and shape of the data.

2.5.1 Data Visualization

To understand data, graph visualization is must. Data visualization helps to understand data. It helps to present information and data to customers. Some of the graphs that are used in univariate data analysis are bar charts, histograms, frequency polygons and pie charts.

The advantages of the graphs are presentation of data, summarization of data, description of data, exploration of data, and to make comparisons of data. Let us consider some forms of graphs now:

Bar Chart A Bar chart (or Bar graph) is used to display the frequency distribution for variables. Bar charts are used to illustrate discrete data. The charts can also help to explain the counts of nominal data. It also helps in comparing the frequency of different groups.

The bar chart for students' marks {45, 60, 60, 80, 85} with Student ID = {1, 2, 3, 4, 5} is shown below in Figure 2.3.

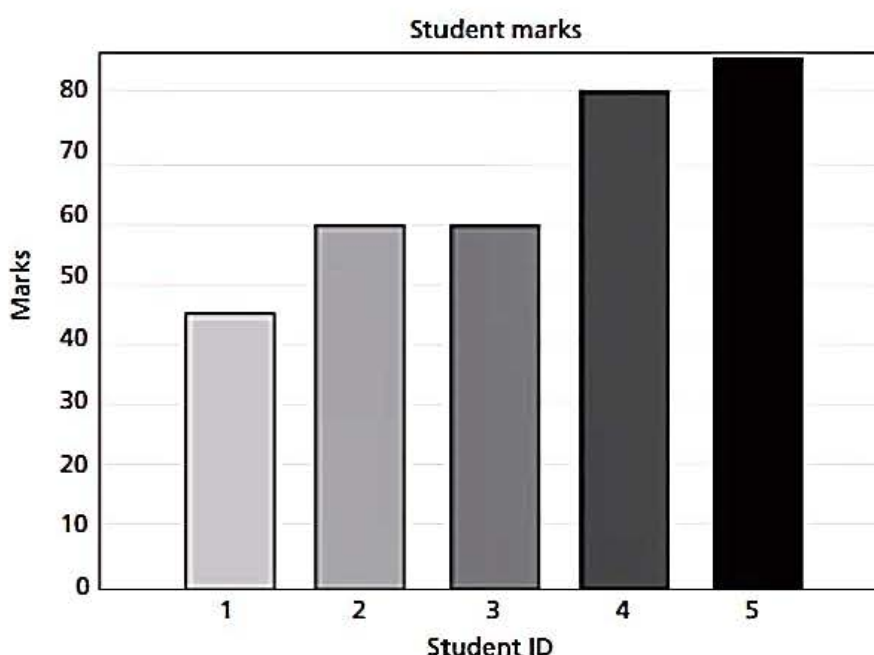


Figure 2.3: Bar Chart

Pie Chart These are equally helpful in illustrating the univariate data. The percentage frequency distribution of students' marks {22, 22, 40, 40, 70, 70, 70, 85, 90, 90} is below in Figure 2.4.

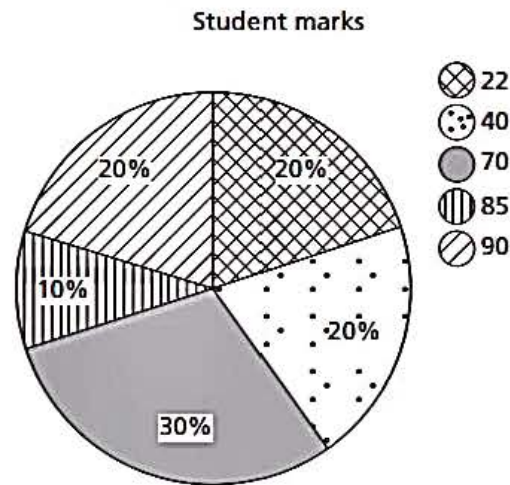


Figure 2.4: Pie Chart

It can be observed that the number of students with 22 marks are 2. The total number of students are 10. So, $2/10 \times 100 = 20\%$ space in a pie of 100% is allotted for marks 22 in Figure 2.4.

Histogram It plays an important role in data mining for showing frequency distributions. The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0–25, 26–50, 51–75, 76–100 is given below in Figure 2.5. One can visually inspect from Figure 2.5 that the number of students in the range 76–100 is 2.

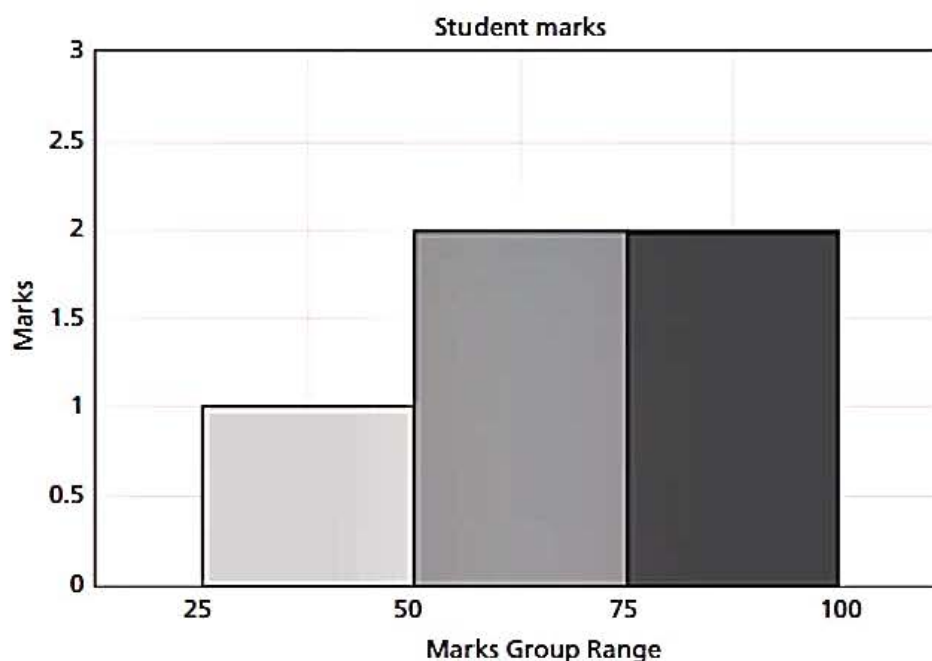


Figure 2.5: Sample Histogram of English Marks

Histogram conveys useful information like nature of data and its mode. Mode indicates the peak of dataset. In other words, histograms can be used as charts to show frequency, skewness present in the data, and shape.

Dot Plots These are similar to bar charts. They are less clustered as compared to bar charts, as they illustrate the bars only with single points. The dot plot of English marks for five students with ID as {1, 2, 3, 4, 5} and marks {45, 60, 60, 80, 85} is given in Figure 2.6. The advantage is that by visual inspection one can find out who got more marks.

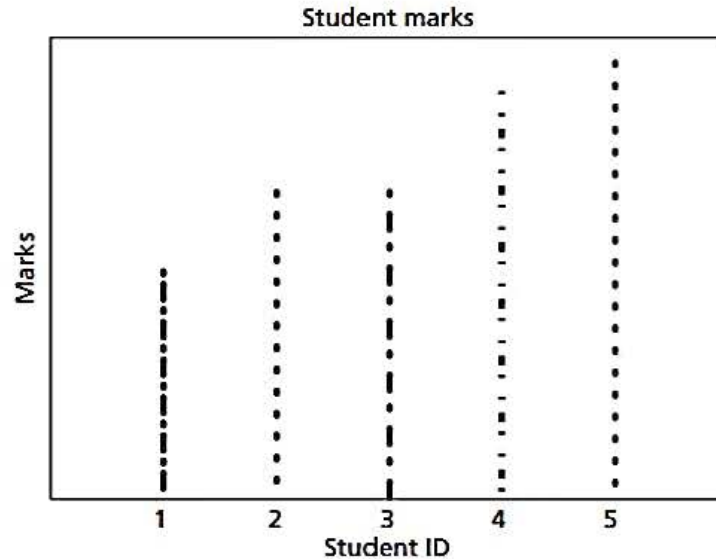


Figure 2.6: Dot Plots

2.5.2 Central Tendency

One cannot remember all the data. Therefore, a condensation or summary of the data is necessary. This makes the data analysis easy and simple. One such summary is called central tendency. Thus, central tendency can explain the characteristics of data and that further helps in comparison. Mass data have tendency to concentrate at certain values, normally in the central location. It is called measure of central tendency (or averages). This represents the first order of measures. Popular measures are mean, median and mode.

1. **Mean – Arithmetic average (or mean)** is a measure of central tendency that represents the 'center' of the dataset. This is the commonest measure used in our daily conversation such as average income or average traffic. It can be found by adding all the data and dividing the sum by the number of observations. Mathematically, the *average* of all the values in the sample (population) is denoted as $\bar{x}$. Let $x_1, x_2, \dots, x_N$ be a set of 'N' values or observations, then the arithmetic mean is given as:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i \quad (2.3)$$

For example, the mean of the three numbers 10, 20, and 30 is $\frac{10 + 20 + 30}{3} = \frac{60}{3} = 20$

- **Weighted mean** – Unlike arithmetic mean that gives the weightage of all items equally, weighted mean gives different importance to all items as the item importance varies. Hence, different weightage can be given to items.

In case of frequency distribution, mid values of the range are taken for computation. This is illustrated in the following computation.

In weighted mean, the mean is computed by adding the product of proportion and group mean. It is mostly used when the sample sizes are unequal.

- **Geometric mean** – Let $x_1, x_2, \dots, x_N$ be a set of 'N' values or observations. Geometric mean is the N^{th} root of the product of N items. The formula for computing geometric mean is given as follows:

$$\text{Geometric mean} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \dots \times x_N} \quad (2.4)$$

Here, n is the number of items and x_i are values. For example, if the values are 6 and 8, the geometric mean is given as $\sqrt[2]{6 \times 8} = \sqrt{48}$. In larger cases, computing geometric mean is difficult. Hence, it is usually calculated as:

$$\text{Anti-log of } \frac{\log(x_1) + \log(x_2) + \dots + \log(x_N)}{N} \quad (2.5)$$

$$= \text{anti-log } \frac{\sum_{i=1}^n \log(x_i)}{N} \quad (2.6)$$

The problem of mean is its extreme sensitiveness to noise. Even small changes in the input affect the mean drastically. Hence, often the top 2% is chopped off and then the mean is calculated for a larger dataset.

2. **Median** – The middle value in the distribution is called median. If the total number of items in the distribution is odd, then the middle value is called median. If the numbers are even, then the average value of two items in the centre is the median. It can be observed that the median is the value where x_i is divided into two equal halves, with half of the values being lower than the median and half higher than the median. A median class is that class where $(N/2)^{\text{th}}$ item is present.

In the continuous case, the median is given by the formula:

$$\text{Median} = L_1 + \frac{\frac{N}{2} - cf}{f} \times i \quad (2.7)$$

Median class is that class where $N/2^{\text{th}}$ item is present. Here, i is the class interval of the median class and L_1 is the lower limit of median class, f is the frequency of the median class, and cf is the cumulative frequency of all classes preceding median.

3. **Mode** – Mode is the value that occurs more frequently in the dataset. In other words, the value that has the highest frequency is called mode. Mode is only for discrete data and is not applicable for continuous data as there are no repeated values in continuous data.

The procedure for finding the mode is to calculate the frequencies for all the values in the data, and mode is the value (or values) with the highest frequency. Normally, the dataset is classified as unimodal, bimodal and trimodal with modes 1, 2 and 3, respectively.

2.5.3 Dispersion

The spreadout of a set of data around the central tendency (mean, median or mode) is called dispersion. Dispersion is represented by various ways such as range, variance, standard deviation, and standard error. These are second order measures. The most common measures of the dispersion data are listed below:

Range Range is the difference between the maximum and minimum of values of the given list of data.

Standard Deviation The mean does not convey much more than a middle point. For example, the following datasets {10, 20, 30} and {10, 50, 0} both have a mean of 20. The difference between these two sets is the spread of data.

Standard deviation is the average distance from the mean of the dataset to each point. The formula for sample standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}} \quad (2.8)$$

Here, N is the size of the population, x_i is observation or value from the population and μ is the population mean. Often, $N - 1$ is used instead of N in the denominator of Eq. (2.8). The reason is that for larger real-world, the division by $N - 1$ gives an answer closer to the actual value.

Quartiles and Inter Quartile Range It is sometimes convenient to subdivide the dataset using coordinates. Percentiles are about data that are less than the coordinates by some percentage of the total value. k^{th} percentile is the property that the $k\%$ of the data lies at or below X_k . For example, median is 50th percentile and can be denoted as $Q_{0.50}$. The 25th percentile is called first quartile (Q_1) and the 75th percentile is called third quartile (Q_3).

Another measure that is useful to measure dispersion is Inter Quartile Range (IQR). The IQR is the difference between Q_3 and Q_1 .

$$\text{Interquartile percentile} = Q_3 - Q_1 \quad (2.9)$$

Outliers are normally the values falling apart at least by the amount $1.5 \times \text{IQR}$ above the third quartile or below the first quartile.

$$\text{Interquartile is defined by } Q_{0.75} - Q_{0.25}. \quad (2.10)$$

Example 2.4: For patients' age list {12, 14, 19, 22, 24, 26, 28, 31, 34}, find the IQR.

Solution: The median is in the fifth position. In this case, 24 is the median. The first quartile is median of the scores below the mean i.e., {12, 14, 19, 22}. Hence, it's the median of the list below 24. In this case, the median is the average of the second and third values, that is, $Q_{0.25} = 16.5$. Similarly, the third quartile is the median of the values above the median, that is {26, 28, 31, 34}. So, $Q_{0.75}$ is the average of the seventh and eighth score. In this case, it is $28 + 31/2 = 59/2 = 29.5$.

Hence, the IQR using Eq. (2.10) is:

$$\begin{aligned} &= Q_{0.75} - Q_{0.25} \\ &= 29.5 - 16.5 = 13 \end{aligned}$$

The half of IQR is called semi-quartile range. The Semi Inter Quartile Range (SIQR) is given as:

$$\begin{aligned}\text{SIQR} &= \frac{1}{2} \times \text{IQR} \\ &= \frac{1}{2} \times 13 = 6.5\end{aligned}\quad (2.11)$$

Five-point Summary and Box Plots The median, quartiles Q_1 and Q_3 , and minimum and maximum written in the order $\langle \text{Minimum}, Q_1, \text{Median}, Q_3, \text{Maximum} \rangle$ is known as five-point summary.

Box plots are suitable for continuous variables and a nominal variable. Box plots can be used to illustrate data distributions and summary of data. It is the popular way for plotting five number summaries. A Box plot is also known as a Box and whisker plot.

The box contains bulk of the data. These data are between first and third quartiles. The line inside the box indicates location – mostly median of the data. If the median is not equidistant, then the data is skewed. The whiskers that project from the ends of the box indicate the spread of the tails and the maximum and minimum of the data value.

Example 2.5: Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9}.

Solution: The minimum is 2 and the maximum is 13. The Q_1 , Q_2 and Q_3 are 3, 8 and 11, respectively. Hence, 5-point summary is {2, 3, 8, 11, 13}, that is, {minimum, Q_1 , median, Q_3 , maximum}.

Box plots are useful for describing 5-point summary. The Box plot for the set is given in Figure 2.7.

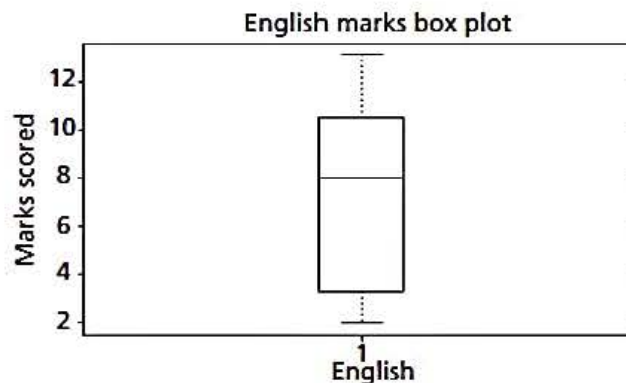


Figure 2.7: Box Plot for English Marks

2.5.4 Shape

Skewness and Kurtosis (called moments) indicate the symmetry/asymmetry and peak location of the dataset.

Skewness

The measures of direction and degree of symmetry are called measures of third order. Ideally, skewness should be zero as in ideal normal distribution. More often, the given dataset may not have perfect symmetry (consider the following Figure 2.8).

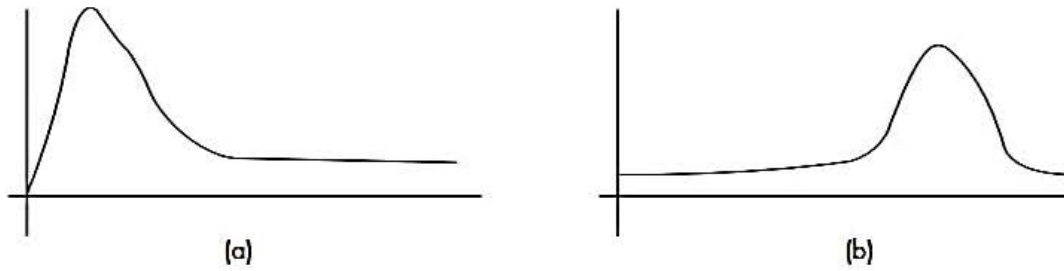


Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

The dataset may also either have very high values or extremely low values. If the dataset has far higher values, then it is said to be skewed to the right. On the other hand, if the dataset has far more low values then it is said to be skewed towards left. If the tail is longer on the left-hand side and hump on the right-hand side, it is called positive skew. Otherwise, it is called negative skew.

The given dataset may have an equal distribution of data. The implication of this is that if the data is skewed, then there is a greater chance of outliers in the dataset. This affects the mean and median. Hence, this may affect the performance of the data mining algorithm. A perfect symmetry means the skewness is zero. In the case of skew, the median is greater than the mean. In positive skew, the mean is greater than the median.

Generally, for negatively skewed distribution, the median is more than the mean. The relationship between skew and the relative size of the mean and median can be summarized by a convenient numerical skew index known as Pearson 2 skewness coefficient.

$$\frac{3 \times (\mu - \text{median})}{\sigma} \quad (2.12)$$

Also, the following measure is more commonly used to measure skewness. Let $X_1, X_2, \dots, X_N$ be a set of ' N ' values or observations then the skewness can be given as:

$$\frac{1}{N} \times \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3} \quad (2.13)$$

Here, μ is the population mean and σ is the population standard deviation of the univariate data. Sometimes, for bias correction instead of N , $N - 1$ is used.

Kurtosis

Kurtosis also indicates the peaks of data. If the data is high peak, then it indicates higher kurtosis and vice versa.

Kurtosis is the measure of whether the data is heavy tailed or light tailed relative to normal distribution. It can be observed that normal distribution has bell-shaped curve with no long tails. Low kurtosis tends to have light tails. The implication is that there is no outlier data. Let $x_1, x_2, \dots, x_N$ be a set of ' N ' values or observations. Then, kurtosis is measured using the formula given below:

$$\frac{\sum_{i=1}^N (x_i - \bar{x})^4 / N}{\sigma^4} \quad (2.14)$$

It can be observed that $N - 1$ is used instead of N in the numerator of Eq. (2.14) for bias correction. Here, $\bar{x}$ and σ are the mean and standard deviation of the univariate data, respectively.

Some of the other useful measures for finding the shape of the univariate dataset are mean absolute deviation (MAD) and coefficient of variation (CV).

Mean Absolute Deviation (MAD)

MAD is another dispersion measure and is robust to outliers. Normally, the outlier point is detected by computing the deviation from median and by dividing it by MAD. Here, the absolute deviation between the data and mean is taken. Thus, the absolute deviation is given as:

$$|x - \mu| \quad (2.15)$$

The sum of the absolute deviations is given as $\sum |x - \mu|$

Therefore, the mean absolute deviation is given as:
$$\frac{\sum |x - \mu|}{N} \quad (2.16)$$

Coefficient of Variation (CV)

Coefficient of variation is used to compare datasets with different units. CV is the ratio of standard deviation and mean, and %CV is the percentage of coefficient of variations.

2.5.5 Special Univariate Plots

The ideal way to check the shape of the dataset is a stem and leaf plot. A stem and leaf plot are a display that help us to know the shape and distribution of the data. In this method, each value is split into a 'stem' and a 'leaf'. The last digit is usually the leaf and digits to the left of the leaf mostly form the stem. For example, marks 45 are divided into stem 4 and leaf 5 in Figure 2.9.

The stem and leaf plot for the English subject marks, say, {45, 60, 60, 80, 85} is given in Figure 2.9.

Stem	Leaf
4	5
5	
6	0 0
7	
8	0 5

Figure 2.9: Stem and Leaf Plot for English Marks

It can be seen from Figure 2.9 that the first column is stem and the second column is leaf. For the given English marks, two students with 60 marks are shown in stem and leaf plot as stem-6 with 2 leaves with 0.

As discussed earlier, the ideal shape of the dataset is a bell-shaped curve. This corresponds to normality. Most of the statistical tests are designed only for normal distribution of data. A Q-Q plot can be used to assess the shape of the dataset. The Q-Q plot is a 2D scatter plot of an univariate data against theoretical normal distribution data or of two datasets – the quartiles of the first and second datasets. The normal Q-Q plot for marks $x = [13 \ 11 \ 2 \ 3 \ 4 \ 8 \ 9]$ is given below in Figure 2.10.

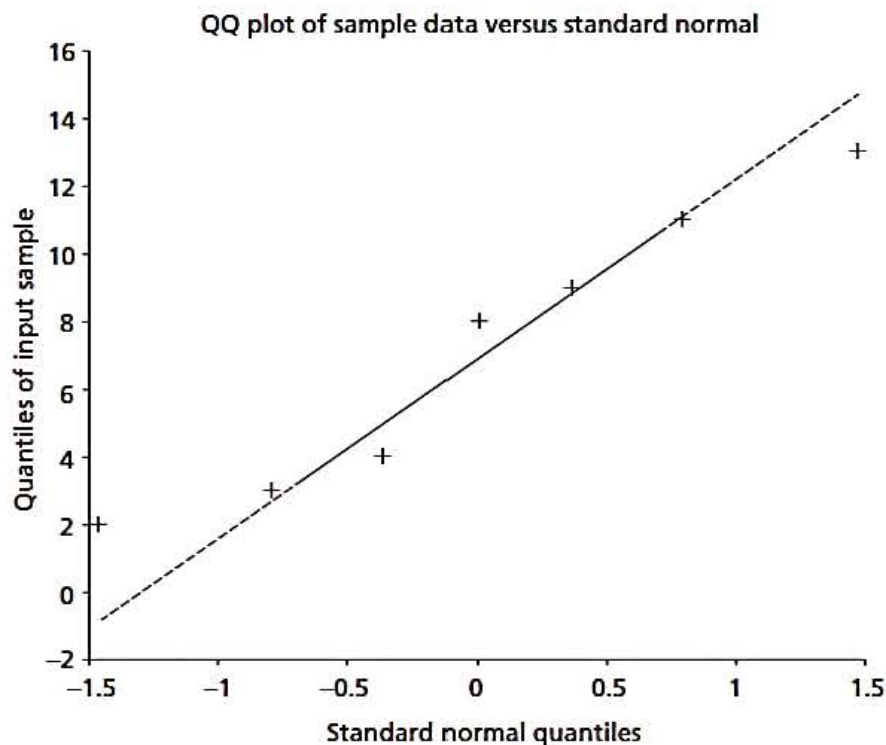


Figure 2.10: Normal Q-Q Plot

Ideally, the points fall along the reference line (45 Degree) if the data follows normal distribution. If the deviation is more, then there is greater evidence that the datasets follow some different distribution, that is, other than the normal distribution shape. In such a case, careful analysis of the statistical investigations should be carried out before interpretation.

This skewness, kurtosis, mean absolute deviation and coefficient of variation help in assessing the univariate data.

2.6 BIVARIATE DATA AND MULTIVARIATE DATA

Bivariate Data involves two variables. Bivariate data deals with causes of relationships. The aim is to find relationships among data. Consider the following Table 2.3, with data of the temperature in a shop and sales of sweaters.

Table 2.3: Temperature in a Shop and Sales Data

Temperature (in centigrade)	Sales of Sweaters (in thousands)
5	200
10	150
15	140
20	75
22	60
23	55
25	20

Here, the aim of bivariate analysis is to find relationships among variables. The relationships can then be used in comparisons, finding causes, and in further explorations. To do that, graphical display of the data is necessary. One such graph method is called scatter plot.

Scatter plot is used to visualize bivariate data. It is useful to plot two variables with or without nominal variables, to illustrate the trends, and also to show differences. It is a plot between explanatory and response variables. It is a 2D graph showing the relationship between two variables.

The scatter plot (Refer Figure 2.11) indicates strength, shape, direction and the presence of Outliers. It is useful in exploratory data before calculating a correlation coefficient or fitting regression curve.

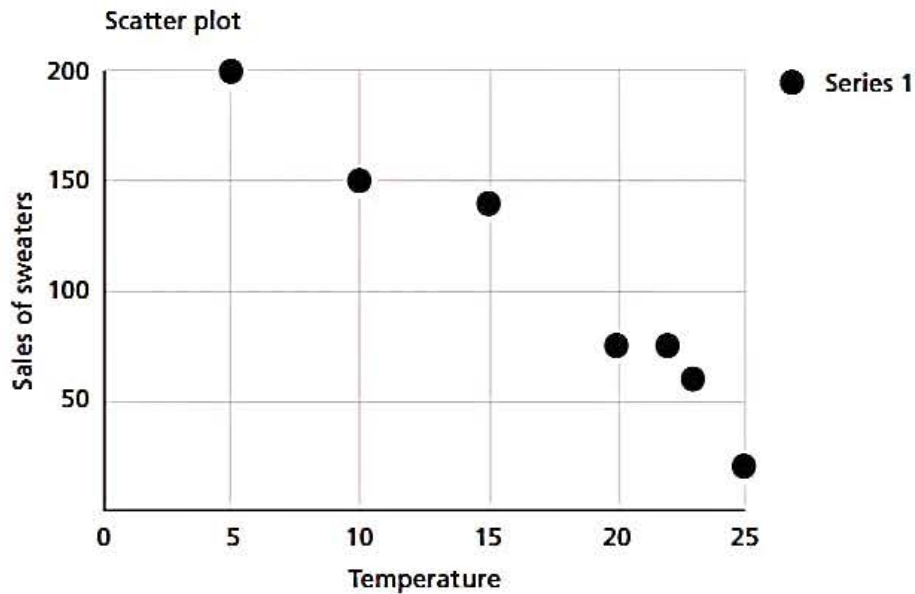


Figure 2.11: Scatter Plot

Line graphs are similar to scatter plots. The Line Chart for sales data is shown in Figure 2.12.

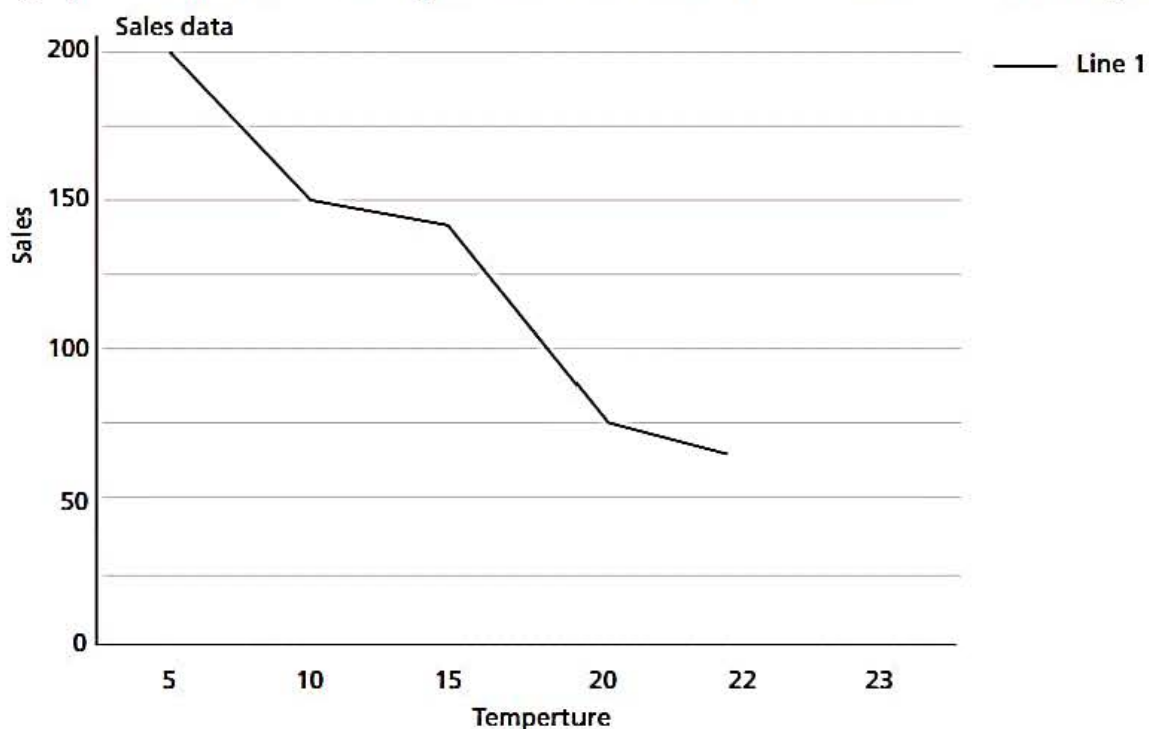


Figure 2.12: Line Chart

2.6.1 Bivariate Statistics

Covariance and Correlation are examples of bivariate statistics. Covariance is a measure of joint probability of random variables, say X and Y . Generally, random variables are represented in capital letters. It is defined as $\text{covariance}(X, Y)$ or $\text{COV}(X, Y)$ and is used to measure the variance between two dimensions. The formula for finding co-variance for specific x , and y are:

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))(y_i - E(Y)) \quad (2.17)$$

Here, x_i and y_i are data values from X and Y . $E(X)$ and $E(Y)$ are the mean values of x_i and y_i . N is the number of given data. Also, the $\text{COV}(X, Y)$ is same as $\text{COV}(Y, X)$.

Example 2.6: Find the covariance of data $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$.

Solution: $\text{Mean}(X) = E(X) = \frac{15}{5} = 3$, $\text{Mean}(Y) = E(Y) = \frac{55}{5} = 11$. The covariance is computed using Eq. (2.17) as:

$$\frac{(1-3)(1-11) + (2-3)(4-11) + (3-3)(9-11) + (4-3)(16-11) + (5-3)(25-11)}{5} = 12$$

The covariance between X and Y is 12. It can be normalized to a value between -1 and $+1$. This is done by dividing it by the correlation of variables. This is called Pearson correlation coefficient. Sometimes, $N - 1$ is also can be used instead of N . In that case, the covariance is $60/4 = 15$.

Correlation

The Pearson correlation coefficient is the most common test for determining any association between two phenomena. It measures the strength and direction of a *linear* relationship between the x and y variables.

The correlation indicates the relationship between dimensions using its sign. The sign is more important than the actual value.

1. If the value is positive, it indicates that the dimensions increase together.
2. If the value is negative, it indicates that while one-dimension increases, the other dimension decreases.
3. If the value is zero, then it indicates that both the dimensions are independent of each other.

If the dimensions are correlated, then it is better to remove one dimension as it is a redundant dimension.

If the given attributes are $X = (x_1, x_2, \dots, x_N)$ and $Y = (y_1, y_2, \dots, y_N)$, then the Pearson correlation coefficient, that is denoted as r , is given as:

$$r = \frac{\text{COV}(X, Y)}{\sigma_X \sigma_Y} \quad (2.18)$$

where, σ_X , σ_Y are the standard deviations of X and Y .

Example 2.7: Find the correlation coefficient of data $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$.

Solution: The mean values of X and Y are $\frac{15}{5} = 3$ and $\frac{55}{5} = 11$. The standard deviations of X and Y are 1.41 and 8.6486, respectively. Therefore, the correlation coefficient is given as ratio of covariance (12 from the previous problem 2.5) and standard deviation of x and y as per Eq. (2.18) as:

$$r = \frac{12}{1.41 \times 8.6486} \approx 0.984$$

2.7 MULTIVARIATE STATISTICS

In machine learning, almost all datasets are multivariable. Multivariate data is the analysis of more than two observable variables, and often, thousands of multiple measurements need to be conducted for one or more subjects.

The multivariate data is like bivariate data but may have more than two dependant variables. Some of the multivariate analysis are regression analysis, principal component analysis, and path analysis.

<i>Id</i>	<i>Attribute 1</i>	<i>Attribute 2</i>	<i>Attribute 3</i>
1	1	4	1
2	2	5	2
3	3	6	1

The mean of multivariate data is a mean vector and the mean of the above three attributes is given as (2, 7.5, 1.33). The variance of multivariate data becomes the covariance matrix. The mean vector is called centroid and variance is called dispersion matrix. This is discussed in the next section.

Multivariate data has three or more variables. The aim of the multivariate analysis is much more. They are regression analysis, factor analysis and multivariate analysis of variance that are explained in the subsequent chapters of this book.

Heatmap

Heatmap is a graphical representation of 2D matrix. It takes a matrix as input and colours it. The darker colours indicate very large values and lighter colours indicate smaller values. The advantage of this method is that humans perceive colours well. So, by colour shaping, larger values can be perceived well. For example, in vehicle traffic data, heavy traffic regions can be differentiated from low traffic regions through heatmap.

In Figure 2.13, patient data highlighting weight and health status is plotted. Here, X-axis is weights and Y-axis is patient counts. The dark colour regions highlight patients' weights vs patient counts in health status.

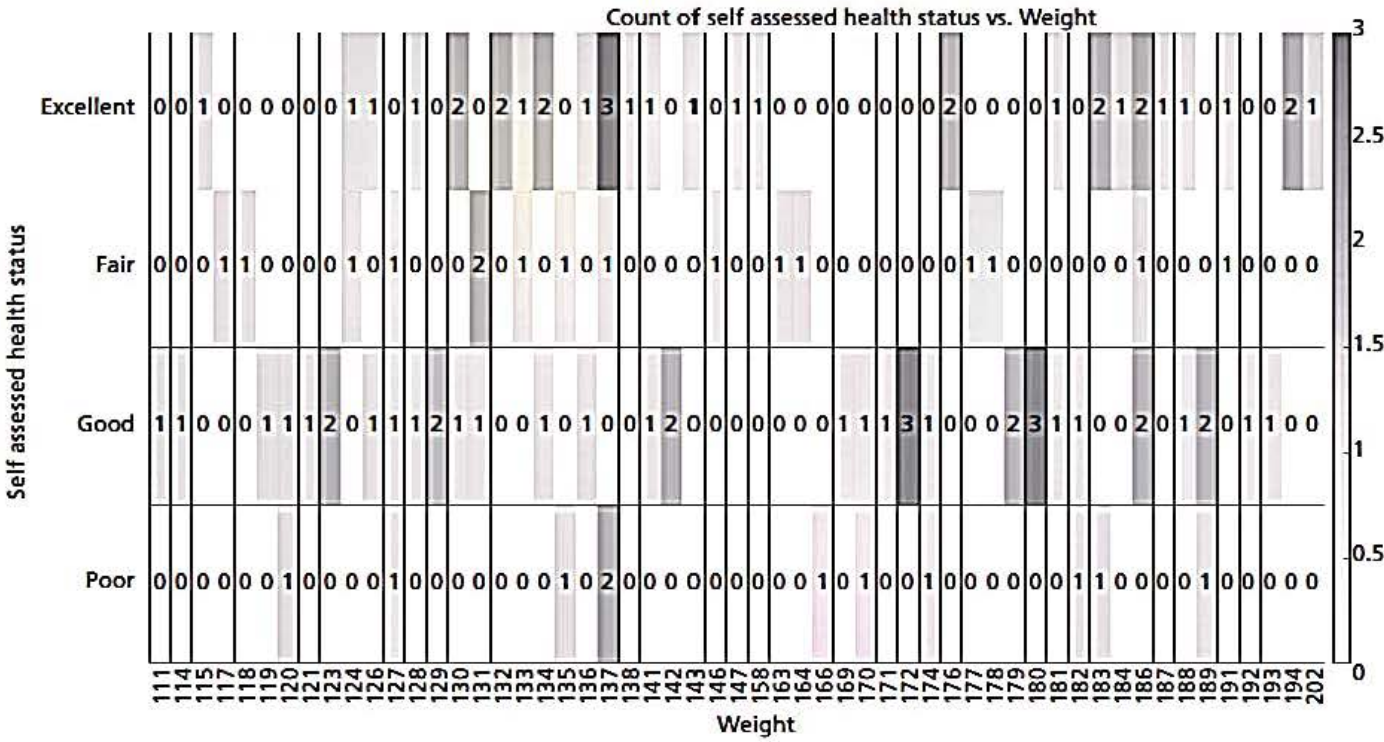


Figure 2.13: Heatmap for Patient Data

Pairplot

Pairplot or scatter matrix is a data visual technique for multivariate data. A scatter matrix consists of several pair-wise scatter plots of variables of the multivariate data. All the results are presented in a matrix format. By visual examination of the chart, one can easily find relationships among the variables such as correlation between the variables.

A random matrix of three columns is chosen and the relationships of the columns is plotted as a pairplot (or scattermatrix) as shown below in Figure 2.14.

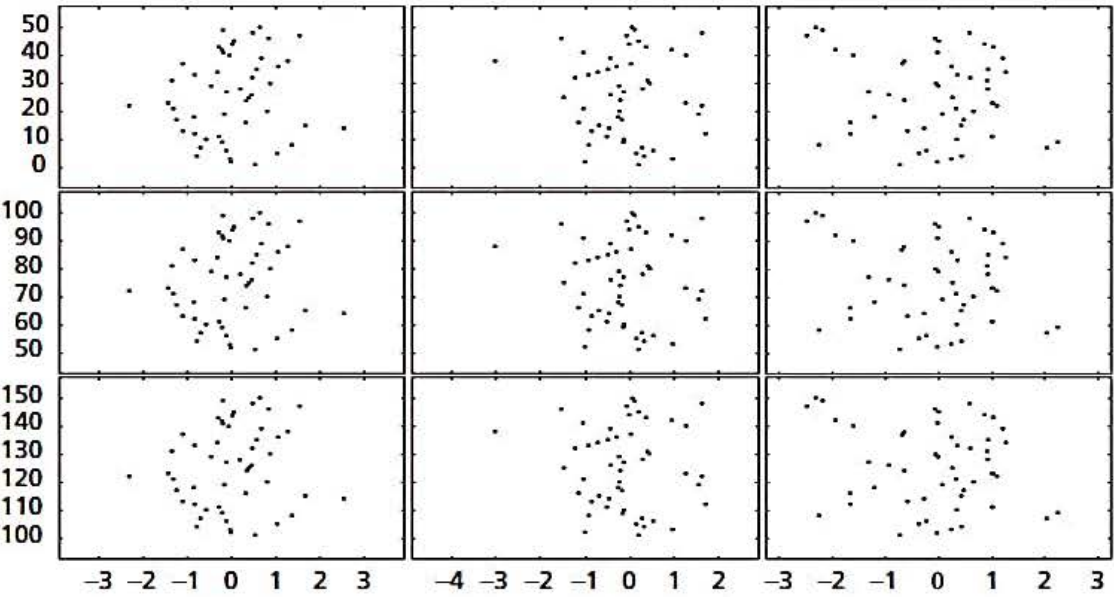


Figure 2.14: Pairplot for Random Data

2.8 ESSENTIAL MATHEMATICS FOR MULTIVARIATE DATA

Machine learning involves many mathematical concepts from the domain of Linear algebra, Statistics, Probability and Information theory. The subsequent sections discuss important aspects of linear algebra and probability.

'Linear Algebra' is a branch of mathematics that is central for many scientific applications and other mathematical subjects. While all branches of mathematics are crucial for machine learning, linear algebra plays a major large role as it is the mathematics of data. Linear algebra deals with linear equations, vectors, matrices, vector spaces and transformations. These are the driving forces of machine learning and machine learning cannot exist without these data types.

Let us discuss some of the important concepts of linear algebra now.

2.8.1 Linear Systems and Gaussian Elimination for Multivariate Data

A linear system of equations is a group of equations with unknown variables.

Let $Ax = y$, then the solution x is given as:

$$x = y/A = A^{-1} y \quad (2.19)$$

This is true if y is not zero and A is not zero. The logic can be extended for N -set of equations with ' n ' unknown variables.

It means if $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix}$ and $y = (y_1 \ y_2 \ \dots \ y_n)$, then the unknown variable x can be computed as:

$$x = y/A = A^{-1} y \quad (2.20)$$

If there is a unique solution, then the system is called consistent independent. If there are various solutions, then the system is called consistent dependant. If there are no solutions and if the equations are contradictory, then the system is called inconsistent.

For solving large number of system of equations, Gaussian elimination can be used. The procedure for applying Gaussian elimination is given as follows:

1. Write the given matrix.
2. Append vector y to the matrix A . This matrix is called augmentation matrix.
3. Keep the element a_{11} as pivot and eliminate all a_{1i} in second row using the matrix operation, $R_2 - \left(\frac{a_{21}}{a_{11}} \right)$, here R_2 is the second row and $\left(\frac{a_{21}}{a_{11}} \right)$ is called the multiplier. The same logic can be used to remove a_{1i} in all other equations.
4. Repeat the same logic and reduce it to reduced echelon form. Then, the unknown variable as:

$$x_n = \frac{y_{nn}}{a_{nn}} \quad (2.21)$$

5. Then, the remaining unknown variables can be found by back-substitution as:

$$x_{n-1} = \frac{y_{n-1} - a_{n-1} \times x_n}{a_{(n-1)(n-1)}} \quad (2.22)$$

This part is called backward substitution.

To facilitate the application of Gaussian elimination method, the following row operations are applied:

1. Swapping the rows
2. Multiplying or dividing a row by a constant
3. Replacing a row by adding or subtracting a multiple of another row to it

These concepts are illustrated in Example 2.8.

Example 2.8: Solve the following set of equations using Gaussian Elimination method.

$$2x_1 + 4x_2 = 6$$

$$4x_1 + 3x_2 = 7$$

Solution: Rewrite this in matrix form as follows:

$$\begin{pmatrix} 2 & 4 & | & 6 \\ 4 & 3 & | & 7 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 4 & | & 6 \\ 4 & 3 & | & 7 \end{pmatrix} R_1 = \frac{R_1}{2}$$

Apply the transformation by dividing the row 1 by 2. There are no general guidelines of row operations other than reducing the given matrix to row echelon form. The operator $\sim$ means reducing to. The above matrix can further be reduced as follows:

$$\sim \begin{pmatrix} 1 & 2 & | & 3 \\ 4 & 3 & | & 7 \end{pmatrix} R_2 = R_2 - 4R_1$$

$$\sim \begin{pmatrix} 1 & 2 & | & 3 \\ 0 & -5 & | & -5 \end{pmatrix} R_2 = R_2 / -5$$

$$\sim \begin{pmatrix} 1 & 2 & | & 3 \\ 0 & 1 & | & 1 \end{pmatrix} R_1 = R_1 - 2R_2$$

$$\sim \begin{pmatrix} 1 & 0 & | & 1 \\ 0 & 1 & | & 1 \end{pmatrix}$$

Therefore, in the reduced echelon form, it can be observed that:

$$x_2 = 1$$

$$x_1 = 1$$

2.8.2 Matrix Decompositions

It is often necessary to reduce a matrix to its constituent parts so that complex matrix operations can be performed. These methods are also known as matrix factorization methods.

The most popular matrix decomposition is called eigen decomposition. It is a way of reducing the matrix into eigen values and eigen vectors.

Then, the matrix A can be decomposed as:

$$A = Q \Lambda Q^T \quad (2.23)$$

where, Q is the matrix of eigen vectors, Λ is the diagonal matrix and Q^T is the transpose of matrix Q .

LU Decomposition

One of the simplest matrix decompositions is LU decomposition where the matrix A can be decomposed matrices:

$$A = LU$$

Here, L is the lower triangular matrix and U is the upper triangular matrix. The decomposition can be done using Gaussian elimination method as discussed in the previous section. First, an identity matrix is augmented to the given matrix. Then, row operations and Gaussian elimination is applied to reduce the given matrix to get matrices L and U .

Example 2.9 illustrates the application of Gaussian elimination to get LU .

Example 2.9: Find LU decomposition of the given matrix:

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{pmatrix}$$

Solution: First, augment an identity matrix and apply Gaussian elimination. The steps are as shown in:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} \quad \boxed{\text{Initial Matrix}}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 3 & 4 & 2 \end{bmatrix} \quad \boxed{R_2 = R_2 - 3R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & -2 & -10 \end{bmatrix} \quad \boxed{R_3 = R_3 - 3R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & -\frac{10}{3} \end{bmatrix} \quad \boxed{R_3 = R_3 - \frac{2}{3}R_2}$$

Now, it can be observed that the first matrix is L as it is the lower triangular matrix whose values are the determiners used in the reduction of equations above such as 3, 3 and $2/3$. The second matrix is U , the upper triangular matrix whose values are the values of the reduced matrix because of Gaussian elimination.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & -\frac{10}{3} \end{pmatrix}$$

It can be cross verified that the multiplication of LU yields the original matrix A . What are the applications of LU decomposition? Some of the applications are finding matrix inverses and determinant. If the order of the matrix is large, then this method can be used.

2.8.3 Machine Learning and Importance of Probability and Statistics

Machine learning is linked with statistics and probability. Like linear algebra, statistics is the heart of machine learning. The importance of statistics needs to be stressed as without statistics; analysis of data is difficult. Probability is especially important for machine learning. Any data can be assumed to be generated by a probability distribution. Machine learning datasets have multiple data that are generated by multiple distributions. So, a knowledge of probability distribution and random variables are must for better understanding of the machine learning concepts.

What is the objective for an experiment? This is about hypothesis or constructing a model. Machine learning has many models. So, the theory of hypothesis, construction of models, evaluating the models using hypothesis testing, significance of the model and evaluation of the model are all linking factors of machine learning and probability theory. Similarly, the construction of datasets using sampling theory is also required.

The subsequent section will deal with the important concepts of statistics and probability.

Probability Distributions

A probability distribution of a variable, say X , summarizes the probability associated with X 's events. Distribution is a parameterized mathematical function. In other words, distribution is a function that describes the relationship between the observations in a sample space.

Consider a set of data. The data is said to follow a distribution if it obeys a mathematical function that characterizes that distribution. The function can be used to calculate the probability of individual observations.

Probability distributions are of two types:

1. Discrete probability distribution
2. Continuous probability distribution

The relationships between the events for a continuous random variable and their probabilities is called a continuous probability distribution. It is summarized as Probability Density Function (PDF). PDF calculates the probability of observing an instance. The plot of PDF shows the shape of the distribution. Cumulative Distributive Function (CDF) computes the probability of an observation $\leq$ value.

Both PDF and CDF are continuous values. The discrete equivalent of PDF in discrete distribution is called Probability Mass Function (PMF).

The probability of an event cannot be detected directly. It should be computed as the area under the curve for a small interval around the specific outcome. This is defined as CDF.

Let us discuss some of the distributions that are encountered in machine learning.

Continuous Probability Distributions Normal, Rectangular, and Exponential distributions fall under this category.

1. **Normal Distribution** – Normal distribution is a continuous probability distribution. This is also known as gaussian distribution or bell-shaped curve distribution. It is the most common distribution function. The shape of this distribution is a typical bell-shaped curve. In normal distribution, data tends to be around a central value with no bias on left or right. The heights of the students, blood pressure of a population, and marks scored in a class can be approximated using normal distribution.

PDF of the normal distribution is given as:

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (2.24)$$

Here, μ is mean and σ is the standard deviation. Normal distribution is characterized by two parameters – mean and variance.

Mostly, one uses the normal distribution curve of mean 0 and a SD of 1. In normal distribution, mean, median and mode are same. The distribution extends from $-\infty$ to $+\infty$. Standard deviation is how the data is spread out.

One important concept associated with normal distribution is z-score. It can be computed as:

$z = \frac{x - \mu}{\sigma}$. When μ is zero and σ is 1, z-score is same as x . This is useful to normalize the data.

Most of the statistical tests expect data to follow normal distribution. To check it, normality tests are used. Normality test of the data can be done by Q-Q plot where CDF of one random variable follows CDF of normal distribution. Then, quantity of one distribution is plotted against other distributions. If they are same, then the plot closely follows the straight line from bottom-left to top-right.

2. **Rectangular Distribution** – This is also known as uniform distribution. It has equal probabilities for all values in the range a, b . The uniform distribution is given as follows:

$$P(X = x) = \begin{cases} \frac{1}{b - a} & \text{for } a \leq x \leq b \\ 0 & \text{Otherwise} \end{cases} \quad (2.25)$$

3. **Exponential Distribution** – This is a continuous uniform distribution. This probability distribution is used to describe the time between events in a Poisson process. Exponential distribution is another special case of Gamma distribution with a fixed parameter of 1. This distribution is helpful in modelling of time until an event occurs.

The PDF is given as follows:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (\lambda > 0) \quad (2.26)$$

Here, x is a random variable and λ is called rate parameter. The mean and standard deviation of exponential distribution is given as β , where, $\beta = \frac{1}{\lambda}$.

Discrete Distribution Binomial, Poisson, and Bernoulli distributions fall under this category.

1. **Binomial Distribution** – Binomial distribution is another distribution that is often encountered in machine learning. It has only two outcomes: success or failure. This is also called Bernoulli trial.

The objective of this distribution is to find probability of getting success k out of n trials. The way to get success out of k out of n number of trials is given as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (2.27)$$

The binomial distribution function is given as follows, where p is the probability of success and probability of failure is $(1 - p)$. The probability of success in a certain number of trials is given as:

$$p^k(1-p)^{n-k} \text{ or } p^k q^{n-k} \quad (2.28)$$

Combining both, one gets PDF of binomial distribution as:

$$\binom{n}{k} p^k (1-p)^{n-k} \quad (2.29)$$

Here, p is the probability of each choice, k is the number of choices, and n is the total number of choices. The mean of binomial distribution is given below:

$$\mu = n \times p \quad (2.30)$$

And the variance is given as:

$$\sigma^2 = np(1-p) \quad (2.31)$$

Hence, the standard deviation is given as:

$$\sigma = \sqrt{np(1-p)} \quad (2.32)$$

2. **Poisson Distribution** – It is another important distribution that is quite useful. Given an interval of time, this distribution is used to model the probability of a given number of events k . The mean rule λ is inclusive of previous events. Some of the examples of Poisson distribution are number of emails received, number of customers visiting a shop and the number of phone calls received by the office.

The PDF of Poisson distribution is given as follows:

$$f(X = x; \lambda) = Pr[X = x] = \frac{e^{-\lambda} \lambda^x}{x!} \quad (2.33)$$

Here, x is the number of times the event occurs and λ is the mean number of times an event occurs.

The mean is the population mean at number of emails received and the standard deviation is $\sqrt{\lambda}$.

3. **Bernoulli Distribution** – This distribution models an experiment whose outcome is binary. The outcome is positive with p and negative with $1 - p$. The PMF of this distribution is given as:

$$f(k; p) = \begin{cases} q = 1 - p & \text{if } k = 0 \\ p & \text{if } k = 1. \end{cases} \quad (2.34)$$

The mean is p and variance is $p(1-p) = q$

Density Estimation

Let there be a set of observed values $x_1, x_2, \dots, x_n$ from a larger set of data whose distribution is not known. Density estimation is the problem of estimating the density function from an observed data. The estimated density function, denoted as, $p(x)$ can be used to value directly for any unknown data, say x_i as $p(x_i)$. If its value is less than ϵ , then x_i is not an outlier or anomaly data. Else, it is categorized as an anomaly data.

There are two types of density estimation methods, namely parametric density estimation and non-parametric density estimation.

Parametric Density Estimation It assumes that the data is from a known probabilistic distribution and can be estimated as $p(x | \Theta)$, where, Θ is the parameter. Maximum likelihood function is a parametric estimation method.

Maximum Likelihood Estimation For a sample of observations, one can estimate the probability distribution. This is called density estimation. Maximum Likelihood Estimation (MLE) is a probabilistic framework that can be used for density estimation. This involves formulating a function called likelihood function which is the conditional probability of observing the observed samples and distribution function with its parameters. For example, if the observations are $X = \{x_1, x_2, \dots, x_n\}$, then density estimation is the problem of choosing a PDF with suitable parameters to describe the data. MLE treats this problem as a search or optimization problem where the probability should be maximized for the joint probabilities of X and its parameter, θ .

For example, this is expressed as $p(X; \theta)$, where, $X = \{x_1, x_2, \dots, x_n\}$

The likelihood of observing the data is given as a function $L(X; \theta)$. The objective of MLE is to maximize this function as $\max L(X; \theta)$.

The joint probability of this problem can be restated as $\prod_{i=1}^n p(x_i; \theta)$.

The computation of the above formula is unstable and the hence the problem is restated as maximum of log conditional probability given θ . This is given as:

$$\sum_{i=1}^n \log p(x_i; \theta) \quad (2.35)$$

Instead of maximizing, one can minimize this function as:

$$\min = -\sum_{i=1}^n \log p(x_i; \theta) \text{ as, often, minimization is preferred over maximization.}$$

This is called negative log-likelihood function.

The relevance of this theory of MLE for machine learning is that MLE can solve the problem of predictive modelling in machine learning. Regression problem is treated in Chapter 5 using the principle of least-square approach. The same problem can be viewed from MLE perspective too. If one assumes that the regression problem can be framed as predicting output y given input x , then for $p(y/x)$, the MLE framework can be applied as:

$$\max \sum \log(y | x_i, h) \quad (2.36)$$

Here, h is the linear regression model. If Gaussian distribution is assumed as it is an obvious fact that most of the data follow Gaussian distribution, then MLE can be stated as:

$$\max \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y_i - h(x_i; \beta)}{2\sigma^2}} \quad (2.37)$$

Here, β is the regression coefficient and x_i is the given sample. One can maximize this function or minimize the negative log likelihood function to provide a solution for linear regression problem. The Eq. (2.37) yields the same answer of the least-square approach.

Stochastic gradient descent (SGD) is an algorithm that is normally used to maximize or minimize any function. This algorithm can be used to minimize the above function to get the results effectively. Hence, many statistical packages use this approach for solving linear regression problem. Theoretically, any model can be used instead of linear regression in the above equation. Logistic regression is another model that can be used in MLE framework. Logistic regression is discussed in Chapter 5 of this book.

Gaussian Mixture Model and Expectation-Maximization (EM) Algorithm In machine learning, clustering is one of the important tasks. It is discussed in Chapter 13. MLE framework is quite useful for designing model-based methods for clustering data. A model is a statistical method and data is assumed to be generated by a distribution model with its parameter, θ . There may be many distributions involved and that is why it is called as mixture model. Since, Gaussian are normally assumed for data, this mixture model is categorized as Gaussian Mixture Model (GMM).

The EM algorithm is one algorithm that is commonly used for estimating the MLE in the presence of latent or missing variables. What is a latent variable? Let us assume that the dataset includes weights of boys and girls. Considering the fact that the boys' weights would be slightly more than the weights of the girls, one can assume that the larger weights are generated by one gaussian distribution with one set of parameters while girls' weights are generated with another set of parameters. There is an influence of gender in the data, but it is not directly present or observable. These are called latent variables. The EM algorithm is effective for estimating the PDF in the presence of latent variables.

Generally, there can be many unspecified distributions with different set of parameters. The EM algorithm has two stages:

1. Expectation (E) Stage – In this stage, the expected PDF and its parameters are estimated for each latent variable.
2. Maximization (M) stage – In this, the parameters are optimized using the MLE function.

This process is iterative, and the iteration is continued till all the latent variables are fitted by probability distributions effectively along with the parameters.

Non-parametric Density Estimation A non-parametric estimation can be generative or discriminative. Parzen window is a generative estimation method that finds $p(x \mid \Theta)$ as conditional density. Discriminative methods directly compute $p(\Theta \mid x)$ as posteriori probability. Parzen window and k -Nearest Neighbour (KNN) rule are examples of non-parametric density estimation. Let us discuss about them now.

Parzen Window Let there be ' n ' samples, $X = \{x_1, x_2, \dots, x_n\}$

The samples are drawn independently, called as identically independent distribution. Let R be the region that covers ' k ' samples of total ' n ' samples. Then, the probability density function is given as:

$$p = k/n \quad (2.38)$$

The estimate is given as:

$$p(x) = \frac{k/n}{V} \quad (2.39)$$

where, V is the volume of the region R . If R is the hypercube centered at x and h is the length of the hypercube, the volume V is h^2 for 2D square cube and h^3 for 3D cube.

The Parzen window is given as follows:

$$\varphi\left(\frac{x_i - x}{h}\right) = \begin{cases} 1 & \text{if } \frac{|x_i - x|}{h} < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.40)$$

The window indicates if the sample is inside the region or not. The Parzen probability density function estimate using Eq. (2.40) is given as:

$$\begin{aligned} p(x) &= \frac{k/n}{V} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{V_n} \varphi\left(\frac{x_i - x}{h}\right) \end{aligned} \quad (2.41)$$

This window can be replaced by any other function too. If Gaussian function is used, then it is called Gaussian density function.

KNN Estimation The KNN estimation is another non-parametric density estimation method. Here, the initial parameter k is determined and based on that k -neighbours are determined. The probability density function estimate is the average of the values that are returned by the neighbours.

2.9 OVERVIEW OF HYPOTHESIS

Data collection alone is not enough. Data must be interpreted to give a conclusion. The conclusion should be a structured outcome. This assumption of the outcome is called a hypothesis. Statistical methods are used to confirm or reject the hypothesis. The assumption of the statistical test is called null hypothesis. It is also called as hypothesis zero (H_0). In other words, hypothesis is the existing belief. The violation of this hypothesis is called first hypothesis (H_1) or hypothesis one. This is the hypothesis the researcher is trying to establish.

There are two types of hypothesis tests, parametric and non-parametric. Parametric tests are based on parameters such as mean and standard deviation. Non-parametric tests are dependent on characteristics such as independence of events or data following certain distribution.

Statistical tests help to:

1. Define null and alternate hypothesis
2. Describe the hypothesis using parameters
3. Identify the statistical test and statistics

4. Decide the criteria called significance value α
5. Compute p -value (probability value)
6. Take the final decision of accepting or rejecting the hypothesis based on the parameters

Hypothesis testing is particularly important as it is an integral part of the learning algorithms. Generally, the data size is small. So, one may have to know whether the hypothesis will work for additional samples and how accurate it is. No matter how effective the statistical tests are, two kinds of errors are involved, that are Type I and Type II.

Type I error is the incorrect rejection of a true null hypothesis and is called false positive. Type II error is the incomplete failure of rejecting a false hypothesis and is called false negative.

During these calculations, one must include the size of the data sample. Degree of freedom indicates the number of independent pieces of information used for the test. It is indicated as v . The mean or variance can be used to indicate the degree of freedom.

Hypothesis Testing

Let us define two important errors called sample error and true (or actual error). Let us assume that D is the unknown distribution, Target function is $f(x)$: $x \in \{0, 1\}$, x is the instance, $h(x)$ is the hypothesis, and sample set is S that derives the samples on instances drawn from X . Then, the actual error is denoted as:

$$\text{error}_D(h) = \text{pr}_{x \in D}\{f(x) \neq h(x)\} \quad (2.42)$$

In other words, true error is the probability that the hypothesis will misclassify an instance that is drawn at random. The point is that population is very large and hence it is not possible to determine true error and can only be estimated. So, another error is called sample error or estimator.

Sample error is with respect to sample S . It is the probability for instances drawn from X , that is, the fractions of S that are misclassified. The sample error is given as follows:

$$\text{error}_S(h) = \frac{1}{n} \sum_{x \in S} \delta(f(x), h(x)) \quad (2.43)$$

Here, $\delta(f(x), h(x))$ is 1 if $f(x) \neq h(x)$, otherwise is zero.

The data associated with the sample has two problems. One is bias and another is variance.

Since the dataset is small in size, the data may be overfitting. To generalize the concept of the population with smaller dataset is difficult. If S is the sample set, then the data is biased and can be estimated as $E[\text{error}_S(h)] - \text{error}_D(h)$. Another data error is variance, that is, deviation of $\text{error}_S(h)$ from $\text{error}_D(h)$. Here, errors are sample error and true or actual error.

p -value

Statistical tests can be performed to either accept or reject the null hypothesis. This is done by the value called p -value or probability value. It indicates the probability of hypothesis being true. The p -value is used to interpret or quantify the test. For example, a statistical test result may give a value of 0.03. Then, one can compare it with the level 0.05. As $0.03 < 0.05$, the result is assumed to be significant. This means that the variables tested are not independent. Here, 0.05 is called significant level. In general, significant level is called α and p -value is compared with α . If $p\text{-value} \leq \alpha$, then the hypothesis H_1 is rejected and if $p\text{-value} > \alpha$, then the hypothesis H_0 is rejected.

Confidence Intervals

The acceptance or rejection of the hypothesis can also be done using confidence interval. The confidence interval is computed as:

$$\text{Confidence interval} = 1 - \text{significant level} \quad (2.44)$$

Confidence level is the range of values that indicates the location of true mean. Confidence intervals indicate the confidence of the result. If the confidence level is 90%, then it infers that there is 90% of chance that the true mean lies in this range and remaining 10% indicates that true mean is not present. For finding this, one requires mean and standard deviation. Then, $\bar{x}$ can be given as

$\bar{x} \pm z \times \frac{s}{\sqrt{N}}$. Here, s is the standard deviation, N is the number of samples, and z is the value associated with 90% and is called % of confidence. This is also called as margin of error.

Sample error is the unbiased estimate of true error. If no information is provided, then both errors are the same. It is, however, often safe to suggest a margin of confidence associated with the hypothesis. The hypothesis with 95% confidence about the sample error can be given as follows:

$$\text{error}_s(h) \pm 1.96 \sqrt{\frac{\text{error}_s(h)(1 - \text{error}_s(h))}{n}} \quad (2.45)$$

This 1.96 indicates the 95% confidence of the error. The number 1.96 can be replaced by any number that is associated with different levels of confidence. The procedure to estimate the difference between two hypothesis, say h_1 and h_2 , is as follows:

1. A parameter d can be chosen to estimate the error of two hypothesis:
2. $d \equiv \text{error}_D(h_1) - \text{error}_D(h_2)$ (2.46)

Here, there are two hypothesis h_1 and h_2 tested on two sample sets s_1 and s_2 . Similarly, n_1 and n_2 are randomly drawn number of samples.

3. The estimator $\hat{d}$ can be estimated as the difference as:

$$\hat{d} \equiv \text{error}_D(h_1) - \text{error}_D(h_2)$$

4. The confidence intervals can be used for the estimator also as follows:

$$\hat{d} \pm z \sqrt{\frac{\text{error}_{s_1}(h_1) - (1 - \text{error}_{s_1}(h_1))}{n_1} + \frac{\text{error}_{s_2}(h_2) - (1 - \text{error}_{s_2}(h_2))}{n_2}} \quad (2.47)$$

Sometimes, it is desirable to find interval L and U such that $N\%$ of the probability falls in this interval.

Let us discuss about comparing two algorithms based on the hypothesis. This is done based on statistical tests. Some of the important statistical tests are discussed in the next section.

2.9.1 Comparing Learning Methods

Some of the methods for comparing the learning programs are given below:

Z-test

Z-test assumes normal distribution of data whose population variation is known. The sample size is assumed to be large. The focus is to test the population mean. The z-statistic is given as:

$$Z = \frac{X - \mu}{\sqrt{\frac{\sigma^2}{n}}} \quad (2.48)$$

Here, X is the input data, and n is number of data elements. μ and σ are mean and standard deviation of X , respectively.

Example 2.10: Let 12 be the population mean (μ) with the population variance (σ^2) of 2. Consider the sample $X = \{1, 2, 3, 4, 5\}$. Apply z-test and show whether the result is significant.

Solution: The sample mean of $X = \frac{15}{5} = 3$ and the number of samples is $n = 5$. Substituting in Eq. (2.48) gives:

$$Z = \frac{X - \mu}{\frac{\sigma}{\sqrt{n}}} = -10.06$$

By checking the critical value at significance 0.05, one can find that the null hypothesis H_0 is rejected.

t-test and Paired *t*-test

t-test is a hypothesis test and checks if the difference between two samples' mean is real or by chance. Here, data is continuous and randomly selected. There will only be small number of samples and variance between groups is real. The *t*-test statistics follows *t*-distribution under null hypothesis and is used when the number of samples < 30 . The distribution follows *t*-distribution rather than Gaussian distribution. It indicates whether two groups are different or not.

One Sample Test In this test, the mean of one group is checked against the set average that can be either theoretical value or population mean. So, the procedure is:

- Select a group
- Compute average
- Compare it with theoretical value and compute *t*-statistic:

$$t = \frac{m - \mu}{\frac{s}{\sqrt{n}}} \quad (2.49)$$

Here, t is *t*-statistic, m is the mean of the group, μ is the theoretical value or population mean, s is the standard deviation, and n is the group size or sample size.

Independent Two Sample *t*-test *t*-statistic for two groups A and B is computed as follows:

$$t = \frac{\text{mean}(A) - \text{mean}(B)}{\sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2} \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}} \quad (2.50)$$

Here, $\text{mean}(A)$ and $\text{mean}(B)$ are for two different samples. N_1 and N_2 are sample sizes of two groups A and B . s^2 is the variance of the two samples and the degree of freedom is given as $N_1 + N_2 - 2$. Then, *t*-statistic is compared with the *t*-critical value.

Let there be one group {8, 6, 7, 6, 3} and another group {7, 8, 8, 8, 2}. The number of elements in each group (n) are 5. For the first group, the mean and variance are 6 and 3.5, respectively. For the second group, the mean and variance are 6.6 and 6.8, respectively. The degree of freedom is number of samples $N - 1 = 4$. The t -statistic formula can be applied for the above data and its value is -0.42 . The critical value is 0.343. Since the value is less than the critical value, one can say the test is insignificant.

Paired t -test It is used to evaluate the hypothesis before and after intervention. The fact is that these samples are not independent. For example, consider the case of an effect of medication for a diabetic patient. The sequence is that first the sugar is tested, then the medication is done, and again the sugar test is conducted to study the effect of medication. In short, in paired t -test, the data is taken from the same subject twice. In an unpaired t -test, the samples are taken independently. In this only one group is involved. The t -statistic is computed as:

$$t = \frac{m - \mu}{\frac{s}{\sqrt{n}}}$$

Here, t is t -statistic, m is the mean of the group, μ is the theoretical value or population mean, s is the standard deviation, and n is the group size or sample size.

Chi-Square Test

Chi-Square test is a non-parametric test. The goodness-of-fit test statistics follows a Chi-Square distribution under null hypothesis and measures the statistical significance between observed frequency and expected frequency, and each observation is independent of each other and follows normal distribution. This is used to compare the observed frequency of some observation (such as, frequency of buying different brands of fans) with the expected frequency of the observation (of buying of each fan by customers) and decide whether any statistical difference exists. This comparison is used to calculate the value of the Chi-Square statistic as:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (2.51)$$

Here, E is the expected frequency, O is the observed frequency and the degree of freedom is $C - 1$, where, C is number of categories. The Chi-Square test allows us to detect the duplication of data and helps to remove the redundancy of values.

Example 2.11: Consider the following Table 2.4, where the machine learning course registration is done by both boys and girls. There are 50 boys and 50 girls in the class and the registration of the course is given in the table. Apply Chi-Square test and find out whether any differences exist between boys and girls for course registration.

Table 2.4: Observed Data

Gender	Registered	Not Registered	Total
Boys	35	15	50
Girls	25	25	50
Total	60	40	100

Solution: Let the null hypothesis be H_0 when there is no difference between boys and girls and H_1 be the alternate hypothesis when there is a significant difference between boys and girls. For applying the Chi-Square test based on the observations, the expectation should be obtained by multiplying the total boys X registered/Total and Total girls X not registered/Total as shown in Table 2.5.

Table 2.5: Expected Data

Gender	Registered	Not Registered	Total
Boys	$\frac{50 \times 60}{100} = 30$	$\frac{50 \times 40}{100} = 20$	50
Girls	$\frac{50 \times 60}{100} = 30$	$\frac{50 \times 40}{100} = 20$	50
Total	60	40	100

The Chi-Statistic is obtained using Eq. (2.51) as follows:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = \frac{(35 - 30)^2}{30} + \frac{(15 - 20)^2}{20} + \frac{(25 - 30)^2}{30} + \frac{(25 - 20)^2}{20} = 4.166 \text{ for degree of}$$

freedom = number of categories – 1 = 2 – 1 = 1. The p value for this statistic is 0.0412. This is less than 0.05. Therefore, the result is significant.

2.10 FEATURE ENGINEERING AND DIMENSIONALITY REDUCTION TECHNIQUES

Features are attributes. Feature engineering is about determining the subset of features that form an important part of the input that improves the performance of the model, be it classification or any other model in machine learning.

Feature engineering deals with two problems – Feature Transformation and Feature Selection. Feature transformation is extraction of features and creating new features that may be helpful in increasing performance. For example, the height and weight may give a new attribute called Body Mass Index (BMI).

Feature subset selection is another important aspect of feature engineering that focuses on selection of features to reduce the time but not at the cost of reliability.

The subset selection reduces the dataset size by removing irrelevant features and constructs a minimum set of attributes for machine learning. If the dataset has n attributes, then time complexity is extremely high as n dimensions need to be processed for the given dataset. For n attributes, there are 2^n possible subsets. If the value of n is high, the problem becomes intractable. This is called 'curse of dimensionality'. Since, as the number of dimensions increases, the time complexity increases. The remedy is that some of the components that do not contribute much can be deleted. This results in the reduction of dimensionality. Choosing optimal attributes becomes a graph search problem. Typically, the feature subset selection problem uses greedy approach by looking for the best choice at the time using locally optimal choice while hoping that it would lead to global optimal solutions.

The features can be removed based on two aspects:

1. **Feature relevancy** – Some features contribute more for classification than other features. For example, a mole on the face can help in face detection than common features like nose. In simple words, the features should be relevant. The relevancy of the features can be determined based on information measures such as mutual information, correlation-based features like correlation coefficient and distance measures. Distance measures are discussed in Chapter 13 of this book.
2. **Feature redundancy** – Some features are redundant. For example, when a database table has a field called Date of birth, then age field is not relevant as age can be computed easily from date of birth. This helps in removing the column age that leads to reduction of dimension one.

So, the procedure is:

1. Generate all possible subsets
2. Evaluate the subsets and model performance
3. Evaluate the results for optimal feature selection

Filter-based selection uses statistical measures for assessing features. In this approach, no learning algorithm is used. Correlation and information gain measures like mutual information and entropy are all examples of this approach.

Wrapper-based methods use classifiers to identify the best features. These are selected and evaluated by the learning algorithms. This procedure is computationally intensive but has superior performance.

Let us discuss some of the important algorithms that fall under this category.

2.10.1 Stepwise Forward Selection

This procedure starts with an empty set of attributes. Every time, an attribute is tested for statistical significance for best quality and is added to the reduced set. This process is continued till a good reduced set of attributes is obtained.

2.10.2 Stepwise Backward Elimination

This procedure starts with a complete set of attributes. At every stage, the procedure removes the worst attribute from the set, leading to the reduced set.

Combined Approach Both forward and reverse methods can be combined so that the procedure can add the best attribute and remove the worst attribute.

2.10.3 Principal Component Analysis

The idea of the principal component analysis (PCA) or KL transform is to transform a given set of measurements to a new set of features so that the features exhibit high information packing properties. This leads to a reduced and compact set of features. Basically, this elimination is made possible because of the information redundancies. This compact representation is of a reduced dimension.

Consider a group of random vectors of the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The *mean vector* of the set of random vectors is defined as:

$$m_x = E\{x\}$$

The operator E refers to the expected value of the population. This is calculated theoretically using the probability density functions (PDF) of the elements x_i and the joint probability density functions between the elements x_i and x_j . From this, the covariance matrix can be calculated as:

$$C = E\{(x - m_x)(x - m_x)^T\} \quad (2.52)$$

For M random vectors, when M is large enough, the mean vector and covariance matrix can be approximately calculated as:

$$m_x = \frac{1}{M} \sum_{k=1}^M x_k \quad (2.53)$$

$$A = \frac{1}{M} \sum_{k=1}^M x_k x_k^T - m_x m_x^T \quad (2.54)$$

This covariance matrix is real and symmetric. If e_i and λ_i (where, $i = 1, 2, \dots, n$) be the set of eigen vectors and corresponding eigen values of the covariance matrix, the eigen values can be arranged in a descending order so that $\lambda_i \geq \lambda_{i+1}$ for $i = 1, 2, \dots, n - 1$. The corresponding eigen vectors are calculated. Based on this, the transform kernel is constructed. Let the transform kernel be A . Then, the matrix rows are formed from the eigen vectors of the covariance matrix.

The mapping of the vectors x to y using the transformation can now be described as:

$$y = A(x - m_x) \quad (2.55)$$

This transform is also called as Karhunen-Loeve or Hotelling transform. The original vector x can now be reconstructed as follows:

$$x = A^T y + m_x \quad (2.56)$$

The goal of PCA is to reduce the set of attributes to a newer, smaller set that captures the variance of the data. The variance is captured by fewer components, which would give the same result as the original, with all the attributes. Here, instead of using all the eigen vectors of the covariance matrix, only a small set that exhibits the variance can be used. By using a small set, the KL transform can achieve maximum compression of the available data.

If K largest eigen values are used, the recovered information would be:

$$x = A_K^T y + m_x \quad (2.57)$$

The advantages of PCA are immense. It reduces the attribute list by eliminating all irrelevant attributes. The PCA algorithm is as follows:

1. The target dataset x is obtained
2. The mean is subtracted from the dataset. Let the mean be m . Thus, the adjusted dataset is $X - m$. The objective of this process is to transform the dataset with zero mean.
3. The covariance of dataset x is obtained. Let it be C .

4. Eigen values and eigen vectors of the covariance matrix are calculated.
5. The eigen vector of the highest eigen value is the principal component of the dataset. The eigen values are arranged in a descending order. The feature vector is formed with these eigen vectors in its columns.

Feature vector = {eigen vector₁, eigen vector₂, ..., eigen vector_n}

6. Obtain the transpose of feature vector. Let it be A .
7. PCA transform is $y = A \times (x - m)$, where x is the input dataset, m is the mean, and A is the transpose of the feature vector.

The original data can be retrieved using the formula given below:

$$\text{Original data } (f) = \{(A)^{-1} \times y\} + m \quad (2.58)$$

$$= \{(A)^T \times y\} + m \quad (2.59)$$

The new data is a dimensionally reduced matrix that represents the original data. Therefore, PCA is effective in removing the attributes that do not contribute. If the original data is required, it can be obtained with no loss of information. Scree plot is a visualization technique to visualize the principal components or variables that play a more important role as compared to other attributes. Scree plot is a visualization technique to visualize the principal components visually. For a randomly selected dataset of 246 attributes, PCA is applied and its scree plot is shown in Figure 2.15. The scree plot indicates that only 6 out of 246 attributes are important.

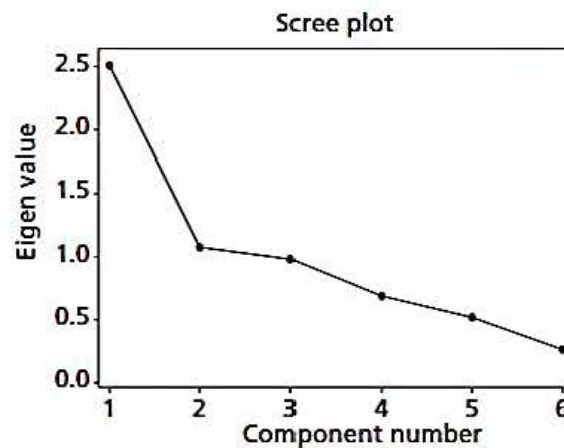


Figure 2.15: Scree Plot

From Figure 2.15, one can infer the relevance of the attributes. The scree plot indicates that the first attribute is more important than all other attributes.

Example 2.12: Let the data points be $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$. Apply PCA and find the transformed data.

Again, apply the inverse and prove that PCA works.

Solution: One can combine two vectors into a matrix as follows:

The mean vector can be computed as Eq. (2.53) as follows:

$$\mu = \begin{pmatrix} \frac{2+1}{2} \\ \frac{6+7}{2} \end{pmatrix} = \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix}$$

As part of PCA, the mean must be subtracted from the data to get the adjusted data:

$$\begin{aligned}x_1 &= \begin{pmatrix} 2 - 1.5 \\ 6 - 6.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} \\x_2 &= \begin{pmatrix} 1 - 1.5 \\ 7 - 6.5 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 0.5 \end{pmatrix}\end{aligned}$$

One can find the covariance for these data vectors. The covariance can be obtained using Eq. (2.54):

$$\begin{aligned}m_1 &= \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} \begin{pmatrix} 0.5 & -0.5 \end{pmatrix} = \begin{pmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{pmatrix} \\m_2 &= \begin{pmatrix} -0.5 \\ 0.5 \end{pmatrix} \begin{pmatrix} -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{pmatrix}\end{aligned}$$

The final covariance matrix is obtained by adding these two matrices as:

$$C = \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix}$$

The eigen values and eigen vectors of matrix C can be obtained (left as an exercise) as $\lambda_1 = 1$, $\lambda_2 = 0$. The eigen vectors are $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The matrix A can be obtained by packing the eigen vector of these eigen values (after sorting it) of matrix C . For this problem, $A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. The transpose of A , $A^T = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ is also the same matrix as it is an orthogonal matrix. The matrix can be normalized by dividing each elements of the vector, by the norm of the vector to get:

$$A = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

One can check that the PCA matrix A is orthogonal. A matrix is orthogonal is $A^{-1} = A$ and $AA^{-1} = I$.

$$\begin{aligned}AA^T &= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\end{aligned}$$

The transformed matrix y using Eq. (2.55) is given as:

$$y = A \times (x - m)$$

Recollect that $(x-m)$ is the adjusted matrix.

$$\begin{aligned}
 y = A(x - m) &= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} \\
 &= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \left(\text{for convenience } 0.5 = \frac{1}{2} \right) \\
 &= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}
 \end{aligned}$$

One can check the original matrix can be retrieved from this matrix as:

$$\begin{aligned}
 x &= A^T y + m \\
 &= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 6 & 7 \end{pmatrix}
 \end{aligned}$$

Therefore, one can infer the original is obtained without any loss of information.

2.10.4 Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is also a feature reduction technique like PCA. The focus of LDA is to project higher dimension data to a line (lower dimension data). LDA is also used to classify the data. Let there be two classes, c_1 and c_2 . Let μ_1 and μ_2 be the mean of the patterns of two classes. The mean of the class c_1 and c_2 can be computed as:

$$\mu_1 = \frac{1}{N_1} \sum_{x_i \in c_1}^n x_i \text{ and } \mu_2 = \frac{1}{N_2} \sum_{x_i \in c_2}^n x_i$$

The aim of LDA is to optimize the function:

$$J(V) = \frac{V^T \sigma_B V}{V^T \sigma_W V} \quad (2.60)$$

where, V is the linear projection and σ_B and σ_W are class scatter matrix and within scatter matrix, respectively. For the two-class problem, these matrices are given as:

$$\sigma_B = N_1(\mu_1 - \mu)(\mu_1 - \mu)^T + N_2(\mu_2 - \mu)(\mu_2 - \mu)^T \quad (2.61)$$

$$\sigma_W = \sum_{x_i \in c_1} (x_i - \mu_1)(x_i - \mu_1)^T + \sum_{x_i \in c_2} (x_i - \mu_2)(x_i - \mu_2)^T \quad (2.62)$$

The maximization of $J(V)$ should satisfy the equation:

$$\sigma_B V = \lambda \sigma_W V \text{ or } \sigma_W^{-1} \sigma_B V = \lambda V \quad (2.63)$$

As $\sigma_B V$ is always in the direction of $(\mu_1 - \mu_2)$, V can be given as:

$$V = \sigma_W^{-1}(\mu_1 - \mu_2) \quad (2.64)$$

Let $V = \{v_1, v_2, \dots, v_d\}$ be the generalized eigen vectors of σ_B and σ_W , where, d is the largest eigen values as in PCA. The transformation of x is then given as:

$$y = V_d^T x \quad (2.65)$$

Like in PCA, the largest eigen values can be retained to have projections.

2.10.5 Singular Value Decomposition

Singular Value Decomposition (SVD) is another useful decomposition technique. Let A be the matrix, then the matrix A can be decomposed as:

$$A = USV^T \quad (2.66)$$

Here, A is the given matrix of dimension $m \times n$, U is the orthogonal matrix whose dimension is $m \times m$, S is the diagonal matrix of dimension $n \times n$, and V is the orthogonal matrix. The procedure for finding decomposition matrix is given as follows:

1. For a given matrix, find AA^T
2. Find eigen values of AA^T
3. Sort the eigen values in a descending order. Pack the eigen vectors as a matrix U .
4. Arrange the square root of the eigen values in diagonal. This matrix is diagonal matrix, S .
5. Find eigen values and eigen vectors for $A^T A$. Find the eigen value and pack the eigen vector as a matrix called V .

Thus, $A = USV^T$. Here, U and V are orthogonal matrices. The columns of U and V are left and right singular values, respectively. SVD is useful in compression, as one can decide to retain only a certain component instead of the original matrix A as:

$$a_{ij} = \sum_{k=1}^n u_{ik} s_k v_{jk} \quad (2.67)$$

Based on the choice of retention, the compression can be controlled.

Example 2.13: Find SVD of the matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}$$

Solution: The first step is to compute:

$$AA^T = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 9 \end{pmatrix} = \begin{pmatrix} 5 & 22 \\ 22 & 97 \end{pmatrix}$$

The eigen value and eigen vector of this matrix can be calculated to get U . The eigen values of this matrix are 0.0098 and 101.9902.

The eigen vectors of this matrix are:

$$u_1 = \begin{pmatrix} 0.2268 \\ 1 \end{pmatrix}$$

$$u_2 = \begin{pmatrix} -4.4086 \\ 1 \end{pmatrix}$$

These vectors are normalized to get the vectors respectively as:

$$u_1 = \begin{pmatrix} 0.2212 \\ 0.9752 \end{pmatrix}$$

$$u_2 = \begin{pmatrix} -0.9752 \\ 0.2212 \end{pmatrix}$$

The matrix U can be obtained by concatenating the above vector as:

$$U = [u_1, u_2] = \begin{pmatrix} 0.2212 & -0.9752 \\ 0.9752 & 0.2212 \end{pmatrix}$$

The matrix V can be obtained by finding $A^T A$. It is $\begin{pmatrix} 17 & 38 \\ 38 & 85 \end{pmatrix}$. The eigen values are 0.0098 and 101.9902. The eigen vectors can be found as follows:

$$v_1 = \begin{pmatrix} 0.447 \\ 1 \end{pmatrix} \text{ when } \lambda = 101.99$$

$$v_2 = \begin{pmatrix} -2.236 \\ 1 \end{pmatrix} \text{ when } \lambda = 0.0098$$

The above can be normalized as follows:

$$v_1 = \begin{pmatrix} 0.4082 \\ 0.9129 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} -0.9129 \\ 0.4082 \end{pmatrix}$$

The matrix V can be obtained by concatenating the above vector as:

$$V = [v_1, v_2] = \begin{pmatrix} 0.4081 & -0.9129 \\ 0.9129 & 0.4082 \end{pmatrix}$$

The matrix S can be found as the diagonal matrix as:

$$S = \begin{pmatrix} \sqrt{101.9902} & 0 \\ 0 & \sqrt{0.0098} \end{pmatrix} = \begin{pmatrix} 10.099 & 0 \\ 0 & 0.099 \end{pmatrix}$$

Therefore, the matrix decomposition $A = U S V^T$ is complete.

The main advantage of SVD is compression. A matrix, say an image, can be decomposed and selectively only certain components can be retained by making all other elements zero. This reduces the contents of image while retaining the quality of the image. SVD is useful in data reduction too.

Summary

1. Data is fact. Data must be converted to information.
2. Data analysis is an operation that converts data to information. Data analytics is a general area encompassing data analysis.
3. Exploratory Data analysis aims to understand the data better.
4. Descriptive statistics aims to summarize the data and data visualization aims to understand the data using charts.
5. Data types are qualitative and quantitative. The qualitative data types include nominal and ordinal. The quantitative data types include interval and ratio.
6. Data also can be classified as univariate data, bivariate data, and multivariate data based on the variables used.
7. The measures of the central tendency include mean, median, mode, and midrange.
8. The most common dispersion measures are range, 5-number summary, interquartile range and standard deviation.
9. Skewness and kurtosis are shape measures. Skewness, kurtosis, mean absolute deviation, and coefficient of variation help in assessing the shape.
10. Visualization is an important aspect of data mining which helps the user to recognize and interpret the results quickly.
11. Bivariate data involves two variables and hence the focus is on finding relationships among two variables.
12. Multivariate data has more variables. The focus is on finding the relations, cause and effects, and statistical inference.
13. Covariance is a measure of joint probability of random variables, say x and y . It is defined as $cov(X, Y)$. Covariance is used to measure variance between two dimensions. Correlation measures the strength of linear relationship among variables.
14. The idea of feature reduction is to transform a given set of measurements to a new set of features, so that the features exhibit high information packing properties. This leads to a reduced and compact set of features.
15. Sampling is a statistical procedure of data reduction Statistics starts by identifying the group for carrying out a study.
16. The assumption of an experiment is called hypothesis. Statistical methods are used to confirm or reject the hypothesis.
17. Z-test, t -tests, and Chi-Square tests are used to evaluate the hypothesis.

Key Terms

Scan for additional 'Key Terms', 'Short Questions' and 'Numerical Problems'



- **Data** – Raw facts.
- **Information** – Processed data.
- **Data Analysis** – A operation that extract knowledge from facts.
- **Data Analytics** – A general area of analysis that encompasses data analysis.
- **Big Data** – A data that has characteristics like volume, velocity, variety and veracity.
- **Data Preprocessing** – A set of operations that are used to condition the data that is needed for machine learning.
- **Noisy Data** – An abnormal data that is different from normal data. Outlier data is same as noisy data but useful.
- **Data Transformations** – Steps that are necessary to convert data to a processable form.
- **Descriptive Statistics** – A branch of statistics that summarizes data.
- **Data Visualization** – A branch of study that deals with the presentation of data.
- **Central Tendency** – It is a summary of data in terms of its center like mean, median and mode.
- **Five Point Summary** – A summary of data in terms of minimum, maximum, mode, and upper/lower quartiles.
- **Kurtosis** – A peak of data.
- **Bivariate Data** – A dataset that has two variables.
- **Pairplot** – A technique of visualizing multivariate data. Multivariate data is one that is characterized by multiple variables.
- **Expectation of X** – Expectation is the mean of random variable observations.
- **Variance** – The average squared difference of each value from the expected value.
- **Covariance** – A measure of the variance between two sets of data.
- **Pearson Correlation Coefficient** – The normalized covariance is called correlation coefficient.
- **Feature Reduction** – Reduction of a set of attributes to a new or simpler set that captures the variance of data.
- **Random Variable** – A set of all possible values of an experiment.
- **Distribution** – Parameterized mathematical function that characterizes the data.
- **Sampling Techniques** – A set of procedures for reducing the data.
- **Hypothesis** – The assumption of testing a data. The current assumption is called null hypothesis and the one that wishes to establish is called alternate hypothesis.
- **Error** – The incorrect rejection of a true null hypothesis (called Type I error) and incorrect rejection of a false hypothesis (called Type II error).

Short Questions

1. What is an univariate data?
2. What are the types of data?

3. Distinguish between 'good' and 'bad' data.
4. What are the problems of data collection?
5. Explain missing data analysis.
6. What are the measures of central tendencies?
7. Why are central tendency and dispersion measures important for data miners?
8. What are the measures of skewness and kurtosis?
9. How interquartile range is useful in eliminating outliers?
10. List out the visualization aids available for exploratory data analysis.
11. What is the use of correlation coefficient for data mining?
12. What are the advantages of LU decomposition?
13. What is the difference between correlation and covariance?
14. What is the need for PCA analysis?
15. Justify the use of scree plots in PCA.
16. Explain the role of probability distributions in machine learning.
17. Explain the basics of binomial distribution.
18. Distinguish between sample error and actual error.
19. List out the types of sampling techniques.
20. What is a hypothesis?

Long Questions

1. Explain in stages the data management life cycle.
2. Explain the types of Big Data with an example.
3. Explain in detail data cleaning processes.
4. Explain in detail univariate data analysis.
5. Explain atleast five charts in detail that help data visualization.
6. Explain the procedure for SVD.
7. Explain the process of obtaining principle components and its relevance in feature reduction.
8. Explain the procedure for hypothesis testing.
9. Explain the procedure for pair- t tests and Chi-Square goodness fit test.

Numerical Problems

1. For a given univariate dataset $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find mean, median, mode, standard deviation and variance.
2. For a given univariate dataset $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find arithmetic mean, geometric mean.

3. For a given univariate dataset $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find five-point summary and plot the box chart.
4. For the given Tables 2.6 and 2.7, perform the descriptive analysis of data:

Table 2.6: Sample Data

Age	Weight
1	4.2
2	4.5
3	4.7
4	5.2
5	6
6	6.2
7	7
8	7.2
9	7.5
10	8.5

Table 2.7: Students Marks Table

Sid	English	Hindi	Maths	Science
1	45	70.5	90	40
2	60	72.5	80	45
3	60	80	90	50
4	80	80	90	80
5	85	72	70	60

- (a) Find *min* and *max* marks scored in each subject.
- (b) Find details of student who scored highest marks in Maths.
- (c) Find the students with marks English >60 and Maths >70.
5. For Univariate Attribute such as Weight, English and Math marks (consider Tables 2.6 and 2.7), find the following:
 - (a) Mean, Median, Mode
 - (b) Weighted Mean, Geometric Mean and Harmonic Mean
 - (c) Variance and Standard Deviation
 - (d) Absolute Deviation, Mean Absolute Deviation, and Median Absolute Deviation
 - (e) Coefficient of Variation
 - (f) Skewness and Kurtosis
 - (g) Five-point summary, IQR, Semi-Quartile
6. For the Bivariate data such as English and Math (consider Tables 2.6 and 2.7), find:
 - (a) Covariance and Correlation between two variables
 - (b) Covariance between English and Hindi Marks

7. Use appropriate Data Visualization to plot the above Table 2.6 using the following charts:

- (a) Bar Plot and Pie Chart
- (b) Histogram, Box Plot, and QQPLOT
- (c) Dot Plot, Line Chart, Scatterplot
- (d) Stem and Leaf Plot

8. Solve the following set of equations using Gaussian elimination method.

$$2x_1 + 5x_2 = 7$$

$$6x_1 + 12x_2 = 18$$

9. Solve the following set of equations using LU decomposition method.

$$2x_1 + 5x_2 = 7$$

$$6x_1 + 12x_2 = 18$$

10. Apply PCA for the following matrix and prove that it works.

$$\begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$$

11. Apply SVD for the following matrix:

$$\begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$$

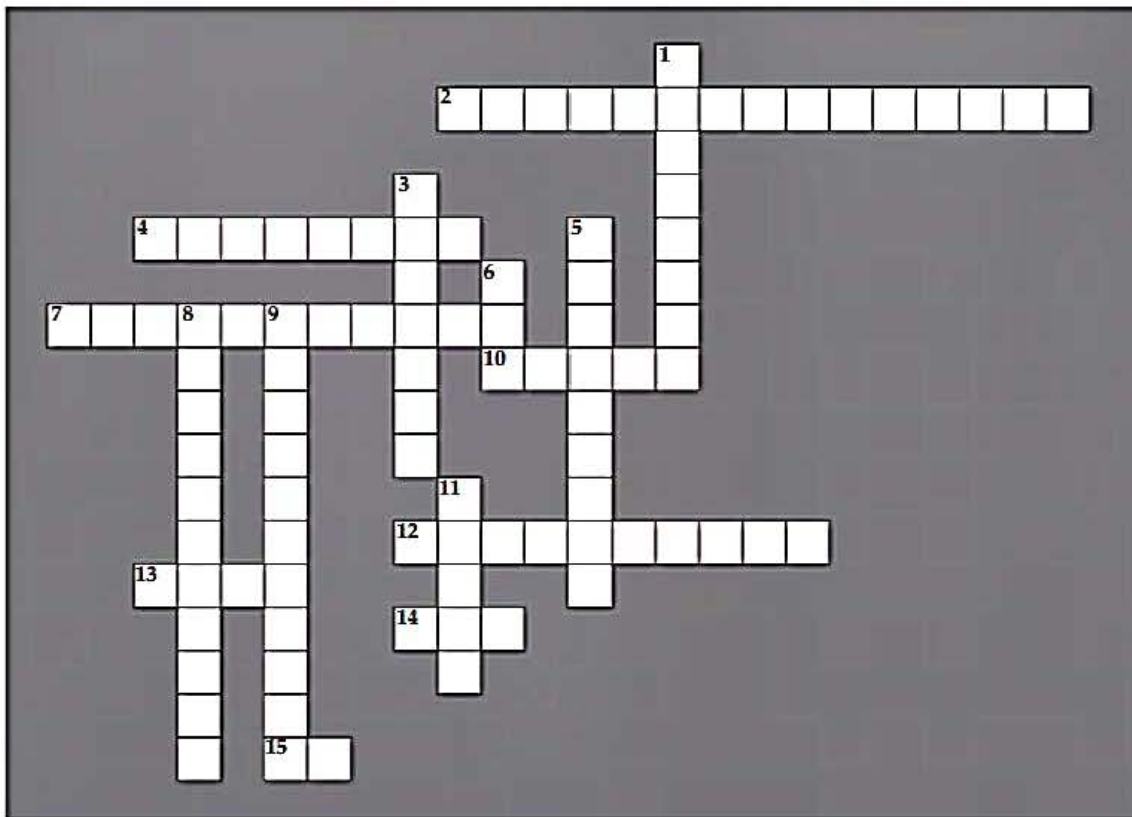
Perform matrix decompositions and prove that SVD works.

12. Find covariance and correlation coefficients for the following two sets of data:

X: 1 2 6 12

Y: 8 12 18 22

Crossword



Across

2. Mean and mode are examples of _____ of univariate data.
4. Noisy data is _____ (Normal/ Abnormal) data.
7. A branch of statistics that is used for summarizing data is called _____ statistics.
10. Kurtosis characterized the _____ of data.
12. The assumption of testing of data is called a _____.
13. Raw facts are called _____.
14. Data wrangling refers to making data suitable for processing. (Yes/No)
15. Pairplot is used to visualize univariate data. (Yes/No)

Down

1. The averaged squared distance from its mean is called _____.
3. The characteristics of Big data are volume, velocity and _____.
5. A dataset of two variables is called _____ data.
6. Visualization helps in presentation of data. (Yes/No)
8. Normalized covariance is called _____.
9. Processed data is _____.
11. Incorrect rejection of true hypothesis is called _____ error.

Word Search

Find and mark the words listed below.

K	V	L	C	Y	X	B	T	Y	P	E	I	Q	S	V	A	R	I	A	N	C	E	I	E	Z
Y	S	G	F	O	I	H	P	K	A	V	E	Z	G	D	T	Y	C	N	T	G	S	N	U	B
U	P	H	O	Y	Y	L	S	I	R	D	T	P	O	A	K	E	E	W	V	J	R	F	M	Z
V	R	J	N	I	E	Y	X	H	A	Z	C	A	V	T	N	S	N	G	Q	C	M	O	J	P
F	E	J	X	A	A	W	U	U	Q	P	X	I	Y	A	L	O	T	D	D	H	I	R	G	L
Y	W	N	F	V	L	D	A	B	N	O	R	M	A	L	I	A	R	E	I	R	P	M	D	Z
E	Y	I	Z	L	Y	J	K	Z	S	J	G	E	T	T	L	U	A	S	S	N	T	A	U	Y
D	B	G	G	W	H	A	N	A	U	F	P	V	A	I	J	T	L	C	C	O	Z	T	I	H
K	R	Q	B	N	Y	I	M	L	H	A	Y	L	P	B	U	X	T	R	M	H	I	I	P	Y
B	A	G	E	I	W	E	P	F	H	Y	E	D	T	W	R	F	E	I	G	A	T	O	T	P
T	V	R	U	A	V	W	S	S	U	R	T	U	B	V	W	B	N	P	I	P	H	N	K	O
C	P	X	Z	C	A	A	C	J	R	V	C	A	H	A	G	F	D	T	N	Z	D	I	O	T
B	U	S	A	O	C	K	R	O	U	E	K	X	G	R	M	K	E	I	H	E	M	E	L	H
Z	L	B	Y	H	N	Q	C	I	N	C	V	M	G	I	J	E	N	V	N	V	R	C	A	E
T	M	K	L	B	R	U	N	U	A	P	C	M	W	E	W	V	C	E	G	F	P	F	Y	S
R	B	V	L	I	D	W	W	W	L	T	V	B	S	T	S	D	Y	P	H	G	W	F	X	I
R	B	D	N	K	R	R	G	B	J	A	E	V	X	Y	G	H	M	V	A	J	X	D	I	S

Data	Information	Variety	Abnormal	Yes
Descriptive	Yes	Central tendency	Shape	Bivariate
No	Variance	Correlation	Hypothesis	Type I